Littlewood–Richardson Coefficients: From Puzzles to the Integrability Hierarchy

[To be determined]

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Abstract

Littlewood–Richardson coefficients $c^{\nu}_{\lambda\mu}$ govern the multiplication of Schur functions, the decomposition of tensor products of GL_n representations, and the intersection theory of Grassmannians. We survey the landscape of combinatorial rules for these coefficients—from the classical Littlewood–Richardson rule through Knutson–Tao–Woodward puzzles to integrable lattice models—and organize them into a hierarchy controlled by the Yang–Baxter equation and its degenerations. At the top sits the fully solvable regime (braided monoidal, spectral parameter, R-matrices); in the middle, the quasi-solvable or coboundary regime (cactus group, crystals with commutor); at the bottom, the symmetric monoidal regime (bare crystals, no spectral data). We argue that every known combinatorial rule for $c^{\nu}_{\lambda\mu}$ and its K-theoretic and equivariant generalizations finds a natural home in this categorical phase diagram, and we identify the open problems that remain.

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1 Introduction

The Littlewood-Richardson coefficients $c^{\lambda}_{\mu\nu}$ are among the most studied numbers in algebraic combinatorics. They appear as the structure constants of the ring of symmetric functions,

$$s_\mu \cdot s_\nu \;=\; \sum_{\lambda} c^{\lambda}_{\mu\nu}\, s_\lambda, \tag{1}$$

as tensor product multiplicities of GL_n representations, and as intersection numbers in the Schubert calculus of Grassmannians. What is remarkable is not that they can be computed—they can, in many ways—but that so many *independent* combinatorial interpretations coexist. The Littlewood-Richardson rule counts skew tableaux satisfying a lattice word condition. The Knutson-Tao-Woodward puzzle model [29] tiles a triangle with three species of unit rhombi. Purbhoo's mosaics [34] give a hybrid between the two. Pipe dreams and bumpless pipe dreams compute the same numbers via reduced decompositions in the symmetric group. Gaetz and Gao's interlacing triangular arrays [16] provide yet another model, discovered from the geometry of Coxeter groups. Each of these was found independently, often by different communities—algebraists, combinatorialists, geometers—and the bijections connecting them, while known, have the character of ad hoc constructions. One is left with a question that is more structural than computational: *why do all of these models exist?*

The integrable perspective

The first hint of a deeper explanation came from Zinn-Justin's observation [**?**] that KTW puzzle pieces *are* the Boltzmann weights of an exactly solvable lattice model. More precisely, the three puzzle tiles carry weights that satisfy the *Yang-Baxter equation* (YBE), the fundamental integrability condition for two-dimensional statistical mechanics. This is not a metaphor. The row transfer matrix of the lattice model literally processes the puzzle triangle row by row, and its partition function evaluates to $c^{\lambda}_{\mu\nu}$. The YBE guarantees that the transfer matrices for different rows commute, which is why the boundary data (the partitions λ, μ, ν) determine the count regardless of the internal tiling—exactly as they should, since the LR coefficient depends on the partitions alone.

Bump and Naprienko [11] elevated this observation to a structural principle. They introduced the *Yang-Baxter groupoid*: its objects are combinatorial models (each specified by a choice of boundary conditions and local vertex weights), and its morphisms are R-matrix transformations arising from the YBE. Two models connected by a morphism compute the same structure constants. Within this framework, the "mysterious" bijections between puzzles, tableaux, and pipe dreams become groupoid morphisms —canonical consequences of the Yang-Baxter equation rather than feats of combinatorial ingenuity. At a still deeper level, Garbali and Gunna [19] showed that all vertex models in this story are representations of the Feigin-Odesskii

shuffle algebra, providing a single algebraic substrate from which the entire zoo of models can be derived.

The categorical phase diagram

The Yang-Baxter groupoid explains the horizontal proliferation of models: many combinatorial objects, all computing the same number, related by R-matrix moves. But there is a second, orthogonal direction of variation that the groupoid alone does not capture. The LR coefficients are the *classical* case—they govern the cohomology ring of the Grassmannian. One can deform to K-theory (Grothendieck polynomials), to quantum K-theory, or to spin models (Hall-Littlewood and q-Whittaker polynomials). Each deformation has its own integrable lattice model, its own R-matrix, and its own family of structure constants—yet all satisfy the full Yang-Baxter equation [39, 22]. The deformation axis is orthogonal to the groupoid axis: it changes *which* structure constants are computed, not the mechanism by which models proliferate.

The central contribution of this paper is a *categorical phase diagram* that organises both directions simultaneously. The horizontal axis parameterises the deformation type (classical, K-theoretic, quantum K-theoretic, spin). The vertical axis parameterises the *integrability level*—the categorical structure that governs the symmetry of the model. We identify three levels, governed by the Hecke algebra chain $H_q \to H_0 \to \mathbb{C}[S_n]$:

1. **Braided monoidal** (generic q). The Hecke generators satisfy $(T_i - q)(T_i + 1) = 0$, and the full Yang-Baxter equation holds. Symmetry is governed by the braid group B_n. This is the world of Zinn-Justin's lattice models, Bump-Naprienko's groupoid, and the rich zoo of combinatorial models related by R-matrix moves.

2. **Coboundary monoidal** ($q = 0$). At $q = 0$ the Hecke relation becomes $T_i(T_i + 1) = 0$; equivalently, the Demazure operators $\pi_i = T_i + 1$ satisfy $\pi_i^2 = \pi_i$, and the braiding degenerates. What survives is a *coboundary* structure: the commutor (the categorical analogue of the swap map) still exists, but it no longer satisfies the braid relation—only the weaker *cactus group* relations of Henriques and Kamnitzer [23]. This is the world of column-level integrability and the "quasi-solvable" models of Buciumas and Scrimshaw [10].

3. **Symmetric monoidal** (crystal limit). At the crystal level, q is specialised away entirely, and only the symmetric group S_n acts. The commutor becomes the trivial swap. Crystal bases [27] provide a purely combinatorial framework: Kashiwara operators, crystal graphs, tensor product rules.

Each downward transition forgets something precise. To state what, we recall the *Schur-Weyl decomposition*: for the defining representation V of GL_n, the tensor power decomposes as

$$V^{\otimes k} \;\cong\; \bigoplus_{\lambda \vdash k} V_\lambda \otimes S_\lambda, \tag{2}$$

where V_λ is the irreducible GL_n-module of highest weight λ and S_λ is the Specht module carrying the action of S_k. The two tensor factors—the "physical" space V_λ and the "multiplicity" space S_λ—play sharply different roles in the hierarchy.

The key result (the *Multiplicity Bundle Theorem*, proved in Section 5) states that the coboundary commutor factorises as

$$\sigma_{[i,j]} \;=\; \mathrm{id}_{V_\lambda} \otimes \rho_{S_\lambda}(w_0), \tag{3}$$

where w_0 is the longest element of S_{j-i+1} and ρ_{S_λ} denotes the Specht module representation. The commutor lives entirely in the multiplicity space—it acts as the identity on the "physical" irreducible V_λ and as the longest-element involution on the Specht module S_λ. The crystal

functor, which retains only the combinatorial skeleton of the representation, kills the multiplicity space: since the tensor square $B(\omega_1)^{\otimes 2}$ of the crystal of the defining representation is multiplicity-free (a property special to the defining representation; the general statement requires the Henriques-Kamnitzer crystal commutor theory), every crystal automorphism is forced to be the identity by rigidity. The commutor becomes invisible.

This gives concrete mathematical content to each transition in the hierarchy. The passage from braided to coboundary forgets the *spectral parameter*: the R-matrix $R(u)$, which depends on a continuous parameter u and satisfies the full parametric YBE, degenerates to the commutor σ, which is a single fixed morphism satisfying only the cactus relations. The passage from coboundary to symmetric forgets the *multiplicity bundle*: the Specht module factor in (2), which carries the nontrivial action of σ, is killed by the crystal functor, leaving only the symmetric swap.

A crucial subtlety, clarified by Kamnitzer and Tingley [26]: the coboundary structure is already present at generic q, constructed from Drinfeld's unitarised R-matrix. At $q = 0$, the braiding dies but the coboundary commutor survives. The transition from braided to coboundary is not a creation but an *extraction*—the coboundary structure was always there, hidden inside the braiding, and the $q = 0$ specialisation strips away the braided superstructure to reveal it. Like a zipper pulled open, the debraiding separates two interlocking layers that were always distinct but moved in concert.

Plan of the paper

Section 2 reviews the classical theory of LR coefficients: their definition via symmetric functions, tableaux, and Schubert calculus, and the puzzle and Hopf algebra perspectives that set the stage for integrability. Section 3 introduces the integrable lattice model framework of Zinn-Justin, presenting the Yang-Baxter equation and the transfer matrix formalism. Section 4 develops the Yang-Baxter groupoid of Bump and Naprienko, explaining how classical bijections between combinatorial models arise as groupoid morphisms. Section 5, the heart of the paper, constructs the categorical phase diagram and proves the Multiplicity Bundle Theorem. Section 6 briefly discusses the shuffle algebra substrate underlying the entire framework. Section 7 collects open questions.

The picture that emerges is, we hope, both clarifying and aesthetically satisfying. The many combinatorial models for LR coefficients are not historical accidents. They are shadows of a single algebraic structure—the integrable R-matrix and its degenerations—cast from different angles onto the combinatorial plane. The categorical phase diagram tells us which angles are available, what is preserved at each level, and what is lost in each projection. The coefficients themselves, those nonnegative integers governing the multiplication of Schur functions, sit at the centre like a fixed point: the one invariant that every shadow faithfully records.

2 Classical Littlewood-Richardson Coefficients

The Littlewood-Richardson coefficients $c^{\nu}_{\lambda,\mu}$ are nonnegative integers that appear, independently and without apparent conspiracy, in three different branches of mathematics. That these three appearances yield the *same* number is already a theorem worth contemplating. We begin with the three definitions, then survey the combinatorial models that make the coefficients computable, and finally recast the entire story in the language of Hopf algebras and transfer operators — the doorway to integrability.

Throughout this section, our running example will be the coefficient

$$c^{(3,2,1)}_{(2,1),\,(2,1)} = 2.$$

This is the smallest instance where the coefficient exceeds 1, and it will accompany us into the lattice model of Section 3.

2.1 Definitions and context

A *partition* $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k > 0)$ is a weakly decreasing sequence of positive integers. Its *Young diagram* is the array of left-justified boxes with λ_i boxes in row i; one may picture it as a skyline read from left to right, each row a terrace one step narrower than the last. The *size* $|\lambda| = \sum_i \lambda_i$ is the total number of boxes.

Definition 2.1 (LR coefficients: symmetric functions)**.** Let Λ denote the ring of symmetric functions over $\mathbb{Z}$, with its basis of Schur functions $\{s_\lambda\}$ indexed by partitions. The *Littlewood-Richardson coefficients* $c^\nu_{\lambda,\mu}$ are the structure constants of this basis:

$$s_\lambda \cdot s_\mu = \sum_\nu c^\nu_{\lambda,\mu}\, s_\nu.$$

Since the Schur functions form a $\mathbb{Z}$-basis of Λ, the coefficients are uniquely determined. That they are *nonnegative* is a deep fact, not at all obvious from this definition.

Definition 2.2 (LR coefficients: representation theory)**.** Let V_λ denote the irreducible polynomial representation of $GL_n(\mathbb{C})$ of highest weight λ (equivalently, the Schur module associated to λ). Then

$$V_\lambda \otimes V_\mu \cong \bigoplus_\nu V_\nu^{\oplus\, c^\nu_{\lambda,\mu}}$$

as GL_n-representations. Here the nonnegativity of the coefficients is manifest: $c^\nu_{\lambda,\mu}$ is the multiplicity of the irreducible V_ν in a tensor product, hence a dimension, hence a nonnegative integer.

Definition 2.3 (LR coefficients: Schubert calculus)**.** Let $\mathrm{Gr}(k,n)$ denote the Grassmannian of k-planes in $\mathbb{C}^n$, and let $\sigma_\lambda \in H^*(\mathrm{Gr}(k,n))$ denote the Schubert class associated to a partition λ fitting inside a $k \times (n-k)$ box. Then

$$\sigma_\lambda \cdot \sigma_\mu = \sum_\nu c^\nu_{\lambda,\mu}\, \sigma_\nu \quad \text{in } H^*(\mathrm{Gr}(k,n)).$$

The geometric content: $c^\nu_{\lambda,\mu}$ counts the number of points in a triple intersection of Schubert varieties in general position. Nonnegativity is again manifest — one is counting points.

Remark 2.4. The fact that Definitions 2.1, 2.2, and 2.3 give the same numbers is a consequence of the Borel-Weil theorem and the classical relationship between the representation ring of GL_n and the cohomology of Grassmannians (see [15] for a unified treatment). But the *feeling* of the three definitions is quite different: in Λ the coefficients are algebraic, in representation theory they are multiplicities, and in Schubert calculus they are enumerative-geometric. That a single integer can simultaneously be all three things is what makes the LR coefficient a meeting place for different parts of mathematics.

The coefficients enjoy several structural properties that constrain and enrich their study:

- **Symmetry.** $c^\nu_{\lambda,\mu} = c^\nu_{\mu,\lambda}$, since multiplication in Λ is commutative.
- **Vanishing.** $c^\nu_{\lambda,\mu} = 0$ unless $|\nu| = |\lambda| + |\mu|$ and $\nu \supseteq \lambda$, $\nu \supseteq \mu$ (containment of Young diagrams).
- **Saturation** (Knutson-Tao [28]). $c^{N\nu}_{N\lambda,\,N\mu} > 0$ if and only if $c^\nu_{\lambda,\mu} > 0$. In other words, the *support* of the LR coefficients is a saturated cone. This resolved the longstanding saturation conjecture and plays a key role in the Horn inequalities characterizing eigenvalues of sums of Hermitian matrices.

2.2 Combinatorial models

The nonnegativity of $c^{\nu}_{\lambda,\mu}$ cries out for a combinatorial proof: exhibit a set whose cardinality is $c^{\nu}_{\lambda,\mu}$. In fact, several such sets are known, each illuminating a different aspect of the coefficient.

Skew tableaux and the Littlewood-Richardson rule

Given partitions $\lambda \subseteq \nu$ (meaning $\lambda_i \leq \nu_i$ for all i), the *skew shape* ν/λ is the set of boxes in ν but not in λ — a diagram with a notch cut from its northwest corner.

Definition 2.5 (Semistandard Young tableau)**.** A *semistandard Young tableau* (SSYT) of shape ν/λ and *content* $\mu = (\mu_1, \mu_2, \ldots)$ is a filling of the boxes of ν/λ with positive integers such that:

(i) entries weakly increase along each row (left to right),

(ii) entries strictly increase down each column,

(iii) the number of entries equal to i is μ_i, for each i.

Definition 2.6 (Lattice word)**.** The *reverse reading word* of a tableau T is obtained by reading the entries of T from right to left in each row, starting from the top row. A word $w_1 w_2 \cdots w_N$ is a *lattice word* (or *ballot sequence*) if, for every prefix $w_1 \cdots w_j$ and every letter i, the number of occurrences of i in the prefix is at least the number of occurrences of $i+1$.

Theorem 2.7 (Littlewood-Richardson rule; see [15, Ch. 5], [37, Ch. 7])**.** *The coefficient $c^{\nu}_{\lambda,\mu}$ equals the number of semistandard Young tableaux of skew shape ν/λ, content μ, whose reverse reading word is a lattice word. Such tableaux are called* LR tableaux.

Example 2.8 (The coefficient $c^{(3,2,1)}_{(2,1),(2,1)} = 2$)**.** The skew shape $(3,2,1)/(2,1)$ consists of three boxes: position $(1,3)$ in the first row, position $(2,2)$ in the second row, and position $(3,1)$ in the third row. We seek fillings with content $(2,1)$ — two 1s and one 2 — satisfying the semistandard and lattice word conditions.

The two LR tableaux are:

$$(1,1,2) \qquad \text{and} \qquad (1,2,1)$$

where denotes a box belonging to $\lambda = (2,1)$ (shaded or omitted in the skew diagram). In the first tableau the reverse reading word is $1,1,2$ and in the second it is $1,2,1$; both are lattice words. The filling (2, 1, 1) with reverse reading word $2,1,1$ is *not* a lattice word (the prefix 2 has more 2s than 1s), so it is excluded. Hence $c^{(3,2,1)}_{(2,1),(2,1)} = 2$.

Knutson-Tao-Woodward puzzles

An entirely different model for the same coefficient was introduced by Knutson, Tao, and Woodward [29].

Definition 2.9 (KTW puzzle)**.** Fix a positive integer N. An *N-puzzle* is a tiling of an equilateral triangle of side length N by the following unit pieces, each carrying labels 0 or 1 on its edges:

1. A triangle of side 1 with all three edges labeled 0.
2. A triangle of side 1 with all three edges labeled 1.
3. A *rhombus* (two unit triangles glued along a shared edge) with edges labeled 0 and 1 in the pattern: the two edges adjacent to one acute angle are labeled 1, and the two edges adjacent to the other acute angle are labeled 0.

The three rhombus orientations (at 0°, 60°, and 120°) are all permitted. In a valid tiling, adjacent pieces must agree on shared edges.

The boundary of the triangle consists of $3N$ edges, which we read as three binary strings of length N: the **south** side (left to right), the **northeast** side (bottom to top), and the **northwest** side (top to bottom). A partition λ fitting inside a $k \times (N-k)$ box is encoded as a binary string of length N via the boundary-path encoding described in Section 2.3 below.

Theorem 2.10 ([29])**.** *For partitions λ, μ, ν fitting inside a $k \times (N-k)$ box, the LR coefficient $c^{\nu}_{\lambda,\mu}$ equals the number of N-puzzles whose south boundary encodes ν, northwest boundary encodes λ, and northeast boundary encodes μ.*

Each valid tiling is a *puzzle witness*: a self-contained combinatorial proof that one unit of $c^{\nu}_{\lambda,\mu}$ exists. For our running example, the two tableaux of Example 2.8 correspond to two puzzle tilings of a triangle of side $N = 5$. We will see in Section 3 that each tiling is a *configuration* of an integrable lattice model — a frozen-in-place snapshot of particle worldlines threading through the triangle.

Remark 2.11. Purbhoo [34] established a direct bijection between KTW puzzles and LR tableaux via an intermediate object called a *mosaic*, making the equivalence of Theorems 2.7 and 2.10 constructive rather than merely numerical.

The multiplicity of models is the paper's motivating mystery. Why should rhombus tilings (a geometric object), lattice-word conditions on skew tableaux (a combinatorial object), and tensor product multiplicities (a representation-theoretic object) all count the same thing? One answer — the one this paper develops — is that all three are shadows of a single algebraic structure: an exactly solvable lattice model whose transfer matrices commute by the Yang-Baxter equation.

2.3 The Hopf algebra perspective

The three definitions of Section 2.1 present LR coefficients as *structure constants* of a product. But the ring of symmetric functions carries more structure: it is a *self-dual graded connected Hopf algebra*, and the LR coefficients appear equally in the coproduct. This dual perspective is less standard in the combinatorics literature, but it is the natural home for transfer operators and the gateway to integrability.

The coproduct

The ring Λ of symmetric functions carries a coproduct

$$\Delta\colon \Lambda \to \Lambda \otimes \Lambda$$

defined on the power-sum generators by $\Delta(p_k) = p_k \otimes 1 + 1 \otimes p_k$ and extended as an algebra homomorphism. On the Schur basis, the coproduct takes the form

$$\Delta(s_\lambda) = \sum_{\mu,\nu} c^{\lambda}_{\mu,\nu}\, s_\mu \otimes s_\nu, \tag{4}$$

where the same LR coefficients appear, now with λ upstairs and the summation over all pairs (μ, ν) with $|\mu| + |\nu| = |\lambda|$. The coproduct encodes *all* LR coefficients for a given λ at once: it "unfolds" the partition like a flower opening, each summand a valid way to split the whole into two interlocking halves.

Self-duality means that the structure constants of the product and coproduct coincide: the coefficient of $s_\mu \otimes s_\nu$ in $\Delta(s_\lambda)$ equals $c^{\lambda}_{\mu,\nu}$, the same integer that appears as the coefficient of s_λ in $s_\mu \cdot s_\nu$. This is the Hopf-algebraic shadow of the fact that the LR coefficients are "the same number" regardless of which definition we use.

Binary string encoding

To make the coproduct computable by transfer operators, we need a finite-dimensional encoding of partitions.

Definition 2.12 (Binary string encoding)**.** Fix a bounding box of height k (number of parts) and width m (largest part). A partition λ with at most k parts, each at most m, is encoded as a binary string $b(\lambda) \in \{0,1\}^{k+m}$ by tracing the northeast boundary of the Young diagram of λ inside the $k \times m$ box, from the southwest corner to the northeast corner:

- a vertical step (north) is recorded as 1,
- a horizontal step (east) is recorded as 0.

The string has exactly k ones and m zeros. For instance, in a 3×4 box, the partition $(3,2,1)$ traces the boundary path $\rightarrow\uparrow\rightarrow\uparrow\rightarrow\uparrow\rightarrow$, giving $b(3,2,1) = 0101010$.

This encoding maps the lattice of partitions inside the $k \times m$ box to a subset of the Boolean lattice $\{0,1\}^{k+m}$. It is also the encoding used for the boundary conditions of KTW puzzles: the binary strings on the three sides of the puzzle triangle are precisely these boundary-path encodings of λ, μ, and ν.

Transfer operators

With partitions encoded as binary strings, we can define operators that act on formal linear combinations of partitions by stripping away horizontal or vertical strips — one row of boxes at a time, interlocking with the previous state like teeth of a zipper.

Definition 2.13 (Free-fermionic transfer operators)**.** Let λ be a partition inside a $k \times m$ box.

1. $T_{\text{free}}(u)\,|\lambda\rangle = \sum_{\substack{\mu \subseteq \lambda \\ \lambda/\mu \text{ h-strip}}} u^{|\lambda/\mu|}\,|\mu\rangle$, where the sum is over all μ such that λ/μ is a *horizontal strip* (at most one box per column). In terms of binary strings, T_{free} slides each 0 (red particle) rightward past one or more 1s (green particles); each crossing contributes a factor of u.
2. $\bar{T}_{\text{free}}(u)\,|\lambda\rangle = \sum_{\substack{\mu \subseteq \lambda \\ \lambda/\mu \text{ v-strip}}} u^{|\lambda/\mu|}\,|\mu\rangle$, where the sum is over all μ such that λ/μ is a *vertical strip* (at most one box per row). In the binary string picture, each 1 (green particle) moves leftward by at most one step past a 0, each move contributing a factor of u.

The key identity connecting these operators to symmetric functions is:

$$s_{\lambda/\mu}(u_1,\ldots,u_n) = \langle \mu \,|\, T_{\text{free}}(u_1)\cdots T_{\text{free}}(u_n)\,|\lambda\rangle, \tag{5}$$

where $\langle\mu|$ denotes the linear functional picking out the coefficient of $|\mu\rangle$. Setting $\mu = \varnothing$ recovers the Schur function $s_\lambda(u_1,\ldots,u_n)$. The analogous formula with $\bar{T}_{\text{free}}$ computes the *conjugate* skew Schur function $s_{\lambda'/\mu'}$.

The coproduct (4) can now be computed by composing transfer operators. Write T_{free} applied n times (with variables $u_1,\ldots,u_n$) to expand $\Delta(s_\lambda)$: one obtains $s_{\lambda/\mu}$ as a polynomial in the u_i, and then decomposes it into Schur functions to read off the LR coefficients.

A computational subtlety

In practice, extracting individual LR coefficients from the transfer operator output requires care. The naïve approach — evaluate the Schur function $s_\nu(1,\ldots,1)$ at various numbers of variables using the hook-content formula, then solve a linear system — fails because the hook-content values are *linearly dependent* across partitions of the same size. For example,

$$\dim_n(3,1) - 3\,\dim_n(2,2) + \dim_n(2,1,1) = 0 \quad \text{for all } n,$$

where $\dim_n(\nu) = s_\nu(1^n)$ is the dimension of the GL_n representation V_ν. So the principal specialization cannot separate all Schur functions of a given degree.

The fix, discovered through computational experiment [**?**], is to use the *Kostka matrix*. Recall that the Kostka number $K_{\nu,\rho}$ counts semistandard Young tableaux of shape ν and content ρ. Ordered by dominance, the Kostka matrix is *unitriangular*: $K_{\nu,\nu} = 1$ and $K_{\nu,\rho} = 0$ unless $\nu \trianglerighteq \rho$ in dominance order. In particular, the Kostka matrix is always invertible, and its inverse can be computed by forward substitution.

One computes the *skew Kostka numbers* $\widetilde{K}_{\lambda/\mu,\,\rho}$ (counting SSYT of skew shape λ/μ and content ρ) via chains of T_{free} applications — each chain of strip removals of prescribed sizes traces a single SSYT. Then, inverting the Kostka matrix recovers the individual LR coefficients $c^\lambda_{\mu,\nu}$ from the skew Kostka numbers via

$$\widetilde{K}_{\lambda/\mu,\,\rho} = \sum_\nu c^\lambda_{\mu,\nu}\, K_{\nu,\rho}.$$

The unitriangularity of K makes this inversion exact over the integers, with no linear dependencies to contend with.

Example 2.14 (Coproduct of $s_{(3,2,1)}$)**.** Applying the transfer operator machinery to $\lambda = (3,2,1)$ yields the coproduct

$$\Delta(s_{(3,2,1)}) = s_{(3,2,1)}\otimes s_\varnothing + s_{(3,1)}\otimes s_{(1,1)} + s_{(2,2)}\otimes s_{(1,1)} + s_{(3,2)}\otimes s_{(1)} + \cdots + 2\, s_{(2,1)}\otimes s_{(2,1)} + \cdots + s_\varnothing \otimes s_{(3,2,1)},$$

with 25 nonzero terms in total. The coefficient 2 at $s_{(2,1)} \otimes s_{(2,1)}$ is our running example.

From transfer operators to lattice models

We close with the observation that motivates the rest of this paper. The operators T_{free} and $\bar{T}_{\text{free}}$ are *row transfer matrices* of a lattice model. Each application of T_{free} processes one row of the lattice: the input state (a binary string encoding a partition) enters from below, the operator produces all valid output states above, weighted by powers of the spectral parameter u. Composing n transfer matrices builds the lattice row by row, and the matrix element $\langle\varnothing|T_{\text{free}}(u_1)\cdots T_{\text{free}}(u_n)|\lambda\rangle$ is a partition function: a sum over all configurations of the lattice with the prescribed boundary conditions.

In Section 3 we will see that this is not merely an analogy. The puzzle pieces of Knutson-Tao-Woodward are precisely the *R-matrix weights* of a free-fermionic vertex model, and the row transfer matrices commute because the R-matrix satisfies the Yang-Baxter equation. The multiplicity of combinatorial models — puzzles, tableaux, mosaics — is then no mystery: they are different ways of summing over the configurations of a single integrable system.

3 The Integrable Lattice Model Perspective

Section 2 closed with an observation: the transfer operators T_{free} and $\bar{T}_{\text{free}}$ are row transfer matrices of a lattice model, and the matrix element $\langle\varnothing|T_{\text{free}}(u_1)\cdots T_{\text{free}}(u_n)|\lambda\rangle$ is a partition function—a weighted sum over all configurations with prescribed boundary conditions. In

this section we make the lattice model explicit. The punchline is that the puzzle pieces of Knutson-Tao-Woodward are *Boltzmann weights* of an exactly solvable vertex model, and the exact solvability follows from a single algebraic identity: the Yang-Baxter equation.

The reader familiar with symmetric functions but not with statistical mechanics should think of a vertex model as follows. One draws a planar graph (the lattice), assigns labels—called *spins*—to the edges, and assigns a *weight* to each vertex as a function of its surrounding spins. A *configuration* is a global assignment of spins to all edges. The *partition function* is the sum of the products of all vertex weights, taken over all configurations with fixed boundary spins. The partition function is the quantity one wants to compute; the Yang-Baxter equation is what makes the computation tractable.

3.1 Zinn-Justin's observation

The starting point is the paper of Zinn-Justin [**?**], which recast the KTW puzzle model in the language of integrable lattice models. We present the key ideas in three steps: reformulation, integrability, and consequences.

Step 1: Puzzles as a vertex model

A KTW puzzle (Definition **??**) tiles an equilateral triangle with unit triangles and rhombi, each edge carrying a label 0 or 1. The internal geometry consists of two families of triangles—upward-pointing ($\triangle$) and downward-pointing (∇)—arranged in rows. Each row processes a horizontal slice of the puzzle triangle, reading edge labels from left to right.

To pass from a tiling model to a vertex model, one performs a standard duality: each *rhombus* (two triangles sharing an internal edge) becomes a *vertex*, and the labels on its four external edges become the spins. The three orientations of rhombus correspond to three orientations of vertex. But a more economical reformulation is available: decompose the row processing into elementary steps, one $\triangle$ or ∇ at a time. Each elementary step reads two input edge labels and produces one output label (for $\triangle$, which is deterministic) or reads one input label and produces two outputs with possible branching (for ∇, which is where all the multiplicity lives). Composing these steps across a row yields the row transfer matrix $T(u)$.

The upshot is that a KTW puzzle of side N becomes a vertex model on an N-row lattice, where each row has $2i-1$ vertices (at row i from the top), the boundary spins are determined by the partitions λ, μ, ν, and each configuration of internal spins is a valid puzzle tiling. The partition function—the sum of the products of all vertex weights—equals the LR coefficient $c^{\nu}_{\lambda,\mu}$.

Remark 3.1. Since all nonzero vertex weights in the KTW model are equal to 1 (the model is *unweighted*), the partition function is simply a *count* of valid configurations. This is why the LR coefficient is a nonnegative integer: it counts puzzle tilings, hence configurations of a vertex model with unit weights.

Step 2: The Yang-Baxter equation

The Yang-Baxter equation (YBE) is the fundamental integrability condition. In its graphical form, it asserts an equivalence between two ways of routing three lines past one another:

$$R_{12}(u-v)\,R_{13}(u)\,R_{23}(v) \;=\; R_{23}(v)\,R_{13}(u)\,R_{12}(u-v), \tag{6}$$

where $R_{ij}(u)$ is a matrix acting on the tensor product of the i-th and j-th copies of the local spin space $\mathbb{C}^2$, and u, v are *spectral parameters*—continuous deformation variables that will play a central role in Section 5. Graphically, the left-hand side represents three lines crossing in the order $(1,2),(1,3),(2,3)$ from right to left, and the right-hand side represents the opposite order

$(2,3),(1,3),(1,2)$. The YBE states that these two composite crossings give the same overall map.

Remark 3.2. The YBE is a system of cubic polynomial equations in the matrix entries of R. For 2×2 spin spaces, R is a 4×4 matrix with 16 entries, and the YBE imposes 64 equations (one for each matrix element of the 8×8 equation (6)). That the six puzzle weights—most of them equal to 1—satisfy this overdetermined system is a genuine surprise, not something one would guess from the puzzle picture. It is this "miracle" that Zinn-Justin's observation brought to light.

Step 3: Transfer matrix commutativity

The YBE has a celebrated consequence: the *commutativity of transfer matrices*. Define the row transfer matrix

$$T(u) \;=\; \mathrm{tr}_a\big(R_{a,1}(u)\, R_{a,2}(u) \cdots R_{a,N}(u)\big), \tag{7}$$

where the trace is over an auxiliary space a and the matrices $R_{a,i}(u)$ act on the tensor product of a with the i-th "physical" spin. The transfer matrix processes one row of the lattice: it reads the spin configuration below and produces the configuration above, summing over the auxiliary spin.

The YBE implies that transfer matrices at different values of the spectral parameter commute:

$$[T(u),\, T(v)] = 0 \qquad \text{for all } u, v. \tag{8}$$

The proof is a direct consequence of (6): one "pushes" the auxiliary line of $T(u)$ past the auxiliary line of $T(v)$ using N applications of the YBE, one at each physical site. The R-matrices appear and cancel in pairs, leaving the two transfer matrices in reversed order.

This is the connection to Section 2.3. The operators $T_{\text{free}}(u)$ and $\bar{T}_{\text{free}}(v)$ defined there are precisely the transfer matrices of the free-fermionic vertex model at spectral parameters u and v. Their commutativity—which we used to compute the coproduct $\Delta(s_\lambda)$—is not an algebraic coincidence. It is the Yang-Baxter equation in action.

What integrability buys

Without the YBE, computing the partition function Z (which equals the LR coefficient) requires enumerating all valid configurations of the lattice model—a sum over exponentially many terms. With the YBE, the situation changes qualitatively.

First, the commuting family $\{T(u)\}_{u\in\mathbb{C}}$ can be *simultaneously diagonalized*. The common eigenvectors are constructed by the *Bethe ansatz*, a technique from quantum inverse scattering that expresses each eigenvector as a product of creation operators applied to a reference state, with the creation parameters (the "Bethe roots") determined by a system of algebraic equations.

Second, the partition function can be expressed as a product of eigenvalues, bypassing the configuration sum entirely. One passes from an *enumerative* computation (count all tilings) to an *algebraic* one (diagonalise a commuting family of matrices). This is the fundamental advantage of integrability: it replaces brute-force enumeration with structure.

For the LR coefficient specifically, the Bethe ansatz machinery yields determinantal formulas, contour integral representations, and functional equations that would be invisible from the puzzle picture alone. The Yang-Baxter equation does not merely *explain* why puzzles work—it gives access to tools that go beyond what any single combinatorial model can offer.

3.2 The R-matrix

We now write down the R-matrix explicitly. A vertex of the model has four edges: two horizontal ("in" from the left and "out" to the right) and two vertical ("in" from below and "out" above).

Each edge carries a spin in $\{0,1\}$. The R-matrix $R\colon \mathbb{C}^2 \otimes \mathbb{C}^2 \to \mathbb{C}^2 \otimes \mathbb{C}^2$ assigns a Boltzmann weight to each combination of input and output spins.

The five-vertex model

For the KTW puzzle model, the nonzero weights are as follows. We write the vertex as $(a,b) \to (c,d)$, where (a,b) are the input spins (left, bottom) and (c,d) are the output spins (right, top):

a	b	c	d	weight
0	0	0	0	1
1	1	1	1	1
0	1	1	0	1
1	0	0	1	1
0	1	0	1	1
1	0	1	0	0

Five of the six possible "ice-type" vertex configurations (those preserving the number of 1s) have weight 1; the sixth—the configuration $(1,0) \to (1,0)$, in which a spin-1 on the left and a spin-0 from below exit unchanged to the right and above—is *forbidden* (weight 0). This is the *five-vertex model*, also known as the free-fermionic model.

In matrix form, writing R in the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$:

$$R \;=\; \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \longleftrightarrow \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{9}$$

where the two forms correspond to two natural orderings of the basis (the right-hand matrix uses the ordering $|00\rangle, |10\rangle, |01\rangle, |11\rangle$ to make the block structure visible). The zero row reflects the forbidden vertex.

The physical interpretation is conservation of "charge" (the number of 1s): every vertex has the same total spin on its input edges as on its output edges. In the puzzle picture, this is the constraint that 1-labeled edges form continuous paths through the triangle.

The spectral parameter

The R-matrix above is the *specialised* form, with all nonzero weights equal to 1. The general five-vertex R-matrix depends on a spectral parameter u:

$$R(u) \;=\; \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & u & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{10}$$

where the weights of the five allowed vertices are $1, 1, 1, u, 1$ in the order listed in the table above (the "straight-through" vertex $(0,1) \to (0,1)$ acquires a weight u, while the crossing vertex $(0,1) \to (1,0)$ retains weight 1). The forbidden vertex $(1,0) \to (1,0)$ remains forbidden at all values of u.

One verifies directly that $R(u)$ satisfies the Yang-Baxter equation (6) with additive spectral parameters:

$$R_{12}(u-v)\, R_{13}(u)\, R_{23}(v) \;=\; R_{23}(v)\, R_{13}(u)\, R_{12}(u-v).$$

The puzzle model corresponds to $u = 0$ (or $u = 1$, depending on normalisation conventions). The spectral parameter does not change *which* configurations are allowed—it changes how they are *weighted*. But because $R(u)$ satisfies the YBE for all u, the transfer matrices $T(u)$ form a commuting family, and all members of the family share the same eigenvectors. Changing u changes the eigenvalues, not the eigenspaces. This is the power of the spectral parameter: a single diagonalisation governs an entire family of models.

Remark 3.3. The spectral parameter u is precisely the variable appearing in the transfer operator $T_{\text{free}}(u)$ of Section 2.3. Setting $u = u_i$ in $R(u)$ and forming the transfer matrix via (7) recovers $T_{\text{free}}(u_i)$ as an operator on binary strings. The formula $s_{\lambda/\mu}(u_1, \ldots, u_n) = \langle \mu | T_{\text{free}}(u_1) \cdots T_{\text{free}}(u_n) | \lambda \rangle$ is then a partition function with *inhomogeneous* spectral parameters, one per row.

Connection to quantum groups

The R-matrix (10) is not an isolated construction. It is the image of the *universal R-matrix* of the quantum group $U_q(\widehat{\mathfrak{sl}}_2)$ in the fundamental (two-dimensional) representation, at a specific value of q corresponding to the free-fermionic point. The Yang-Baxter equation is the defining relation of the quantum group: any solution of the YBE furnishes a representation of $U_q(\widehat{\mathfrak{sl}}_2)$, and conversely, every representation yields an R-matrix satisfying the YBE.

This connection is important for two reasons. First, it places the puzzle model in the vast landscape of quantum group representations, where powerful algebraic tools (Bethe ansatz, fusion, quantum inverse scattering) are available. Second, it explains why R-matrices for *different* symmetric function families (Schur, Grothendieck, Hall-Littlewood) are all solutions of the same equation: they are different representations of the same quantum group, or of closely related quantum groups in the same family. We turn to these extensions next.

3.3 Extensions: the lattice model zoo

The five-vertex R-matrix of Section 3.2 governs the Schur case: its partition function computes LR coefficients, and its transfer matrices build Schur functions row by row. But the Schur functions are only the first member of a hierarchy of symmetric function families, each with its own R-matrix and its own lattice model. The integrable perspective extends to each level of the hierarchy, and the R-matrix changes in systematic, algebraically controlled ways.

Grothendieck polynomials

Wheeler and Zinn-Justin [39] showed that the *stable Grothendieck polynomials*—the K-theoretic analogues of Schur functions, representing Schubert classes in the K-theory of the Grassmannian—arise as partition functions of a *trigonometric* five-vertex model. The R-matrix is again five-vertex (one vertex type forbidden), but the weights depend on an additional parameter β that deforms the Schur case:

$$R^{\text{Groth}}(u;\beta) \;=\; \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & u+\beta & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1+\beta u \end{pmatrix}.$$

At $\beta = 0$ this reduces to the Schur R-matrix (10). The parameter β measures the "K-theoretic deformation": it introduces corrections beyond cohomology, corresponding to higher-order terms in the Grothendieck group. The key fact is that this deformed R-matrix still satisfies the Yang-Baxter equation, so the transfer matrices still commute, and the partition function still computes the structure constants—now the K-theoretic LR coefficients of Buch [**?**].

Hall-Littlewood and spin models

A different direction of generalisation leads to *higher-spin* models. The five-vertex model has spins in $\{0, 1\}$; allowing spins in $\{0, 1, \ldots, s\}$ for $s > 1$ produces lattice models whose partition functions are *Hall-Littlewood polynomials* and their relatives.

Borodin and Wheeler [7] and Wheeler and Zinn-Justin [**?**] developed a family of *coloured* and *higher-rank* vertex models whose partition functions compute spin Hall-Littlewood polynomials. Gunna, Wheeler, and Zinn-Justin [22] showed that the structure constants of these spin polynomials—analogues of LR coefficients for Hall-Littlewood functions—are *positive polynomials in q*. This positivity was not obvious from any previously known formula; it emerged from the lattice model as a natural consequence of the vertex weights being nonnegative.

These higher-spin models are "fully solvable" in the sense of satisfying the complete Yang-Baxter equation, in contrast to the "quasi-solvable" models of the next example.

Demazure characters and the $q = 0$ specialisation

Yang [40] and Brubaker, Bump, Buciumas, and Gustafsson [8] constructed vertex models whose partition functions are *key polynomials* (Demazure characters)—the characters of Demazure modules for GL_n. These models use a coloured five-vertex model at the specialisation $q = 0$ of the Hecke algebra parameter.

At $q = 0$, the Hecke relation $(T_i - q)(T_i + 1) = 0$ degenerates to $T_i(T_i + 1) = 0$, and the Demazure operators $\pi_i = T_i + 1$ become idempotent: $\pi_i^2 = \pi_i$. The R-matrix still satisfies the YBE restricted to pairs of adjacent rows (the "column-level" integrability of Buciumas and Scrimshaw), but the full three-line YBE fails. What survives is a weaker algebraic structure—a *coboundary* category governed by the cactus group rather than the braid group. We will develop this distinction systematically in Section 5.

The states of the Demazure vertex model biject with *Demazure crystal* elements, and the partition function manifestly computes the key polynomial. Since the key polynomial expands positively in the monomial basis, and the Schur function expands positively in key polynomials (via the Kostka numbers), the lattice model provides a direct bridge between the crystal combinatorics of Kashiwara [27] and the integrable lattice models of Zinn-Justin—a bridge that passes through the Hecke algebra at $q = 0$.

The emerging pattern

We can now see a pattern that is too regular to be coincidental. At each level of the character hierarchy—Schur, Grothendieck, key, Hall-Littlewood, Macdonald—there is a solvable lattice model whose partition function computes the corresponding symmetric (or nonsymmetric) function:

Function	**R-matrix type**	**Reference**
Schur	rational 5-vertex	[**?**]
Grothendieck	trigonometric 5-vertex	[39]
Key / Demazure	coloured 5-vertex, $q = 0$	[40, 8]
Hall-Littlewood	higher-spin	[**?**]
Macdonald	elliptic / higher-rank	—

Integrability is not an accident of the Schur case. It is a structural feature of the entire hierarchy of symmetric function theories. Each level has its own R-matrix, its own Yang-Baxter equation (or a controlled degeneration thereof), and its own family of commuting transfer matrices.

This raises the question that drives the remainder of the paper. If every character has a solvable model, what is the *relationship* between these models? When we deform the Schur R-matrix to the Grothendieck R-matrix, or specialise it to the Demazure case, what algebraic structure governs the deformation? Are the bijections between different combinatorial models (puzzles, tableaux, pipe dreams) for the *same* character also controlled by the Yang-Baxter equation?

The answer to the last question is yes, and the organising framework is the *Yang-Baxter groupoid* of Bump and Naprienko—the subject of Section 4. The answer to the deformation question leads to the *categorical phase diagram* of Section 5, where the vertical axis—braided, coboundary, symmetric—classifies the integrability level, and the horizontal axis—rational, trigonometric, elliptic—classifies the deformation type. Together, these two axes organise the full zoo of lattice models into a single coherent picture.

4 The Yang-Baxter Groupoid and the Five-Vertex Classification

Section 3 established that the KTW puzzle model is an integrable vertex model: the puzzle pieces carry R-matrix weights satisfying the Yang-Baxter equation, and the LR coefficient is a partition function. We also saw that the Schur case is not isolated — Grothendieck, Hall-Littlewood, and Demazure characters each have their own solvable (or quasi-solvable) lattice model, each with its own R-matrix. The section closed with two questions: what is the *relationship* between the many combinatorial models for the same coefficient, and what is the *relationship* between the R-matrices at different levels of the character hierarchy?

This section answers both questions. The first answer is the *Yang-Baxter groupoid* of Bump and Naprienko [11]: classical bijections between combinatorial models for LR coefficients are morphisms in a groupoid whose composition law is the Yang-Baxter equation. The second answer is a *classification* of five-vertex R-matrices satisfying the YBE, which organises the R-matrices into a strict three-level hierarchy. Together, these two results set the stage for the categorical phase diagram of Section 5.

4.1 The groupoid framework of Bump and Naprienko

The starting point is a simple observation: if the Yang-Baxter equation (6) holds, then replacing the crossing pattern $R_{12}R_{13}R_{23}$ with $R_{23}R_{13}R_{12}$ in a portion of the lattice does not change the partition function. Each such replacement is a *local bijection* on configurations — it permutes the internal spin assignments without altering the boundary data or the total weight. Bump and Naprienko [11] elevate this observation to a categorical framework: the *Yang-Baxter groupoid*.

Objects and morphisms

The objects of the Yang-Baxter groupoid are *models*: each model is specified by a choice of lattice domain (the shape of the grid), boundary data (the partitions or binary strings fixed on the boundary edges), and local vertex weights (the R-matrix entries at each vertex). Two models may have different lattice domains — a triangular puzzle grid versus a rectangular grid of Gelfand-Tsetlin type, say — but both compute the same structure constants through their respective partition functions.

The morphisms are *sequences of Yang-Baxter moves*. A single YBE move acts on a local patch of the lattice containing three crossing lines: it replaces the crossing order $(1,2),(1,3),(2,3)$ with $(2,3),(1,3),(1,2)$, or vice versa. Because the YBE guarantees that both orderings produce the same map on boundary spins, the move defines a bijection between the configurations of the original local patch and the configurations of the transformed patch, weighted consistently.

A *groupoid morphism* is a finite composition of such local moves, carrying one model to another while preserving the partition function at every step.

The groupoid axioms are satisfied by construction:

- *Identity.* The trivial sequence of zero moves is the identity morphism on any model.
- *Composition.* Composing two sequences of YBE moves gives another sequence.
- *Inverses.* Every YBE move is its own inverse (applying it twice returns to the original crossing order), so every morphism is invertible.

The crucial additional property is *coherence*: if two different sequences of YBE moves relate the same pair of models, they induce the *same* bijection on configurations. This follows from the Zamolodchikov tetrahedron equation in low rank and from braid group relations in general — the braid group acts faithfully on the space of crossing orders, and the YBE is precisely the braid relation. In practice, coherence means that groupoid morphisms are well-defined maps, not merely equivalence classes of move sequences.

Remark 4.1. The groupoid is not a group: different objects (models) are not isomorphic in general — a puzzle model on a triangular lattice and a Gelfand-Tsetlin model on a rectangular lattice have genuinely different configuration spaces. But whenever two models *are* connected by a morphism, the morphism provides a canonical, weight-preserving bijection between their configuration sets.

Recovering classical bijections

The power of the groupoid framework lies in its ability to *explain* bijections that were previously understood only through explicit, often intricate constructions. Bump and Naprienko show that several classical bijections in the theory of LR coefficients are Yang-Baxter groupoid morphisms:

1. **The commutation isomorphism.** The symmetry $c^{\nu}_{\lambda,\mu} = c^{\nu}_{\mu,\lambda}$ is obvious from the Schur function definition ($s_\mu \cdot s_\nu = s_\nu \cdot s_\mu$ by commutativity), but at the level of puzzle tilings it is nontrivial: the two sides correspond to different boundary conditions, and one must construct an explicit bijection between the two sets of tilings. In the groupoid framework, this bijection is a morphism obtained by "sliding" one set of auxiliary lines past another using YBE moves. The commutativity of multiplication becomes a geometric consequence of integrability.

2. **The Schützenberger involution.** Gustafsson and Westerlund [**?**] showed that the Schützenberger involution on semistandard Young tableaux — an involution that plays a central role in the theory of jeu de taquin and crystal bases — arises as a Yang-Baxter groupoid morphism. Their construction interprets the involution as a sequence of R-matrix moves that "reverses the spectral parameters," transforming the lattice model by reflecting the order in which rows are processed. The fact that this reflection preserves the partition function is a direct consequence of transfer matrix commutativity (8).

3. **The puzzle–tableau correspondence.** The bijection between KTW puzzle tilings and LR skew-tableaux, mediated by Purbhoo's mosaics [34], has a natural interpretation as a groupoid morphism. The two models — puzzles on a triangular domain and tableaux on a skew shape — correspond to different objects in the groupoid, and the bijection between them is a composition of YBE moves that deforms the triangular lattice into the rectangular lattice underlying the tableau model. We state this as a guiding principle rather than a proven theorem: verifying it requires checking that each step of Purbhoo's mosaic correspondence can be decomposed into elementary R-matrix moves, a computation that has been carried out for small examples ($n \leq 4$) but not yet in full generality.

The unifying message is this: the "mysterious" bijections of LR coefficient theory are not feats of combinatorial ingenuity. They are canonical consequences of the Yang-Baxter equation — different paths through the groupoid, all connecting the same pair of models, all inducing the same bijection by coherence.

Connection to Gaetz-Gao

The scope of the groupoid extends beyond the classical models. Gaetz and Gao [16] proved that every *integrable triangle-accessible* (ITA) lattice model — a class they define by axiomatising the properties needed for a vertex model to compute LR-type structure constants — bijects canonically with a KTW puzzle model. Their result implies that puzzles are, in a precise sense, a *universal* object in the groupoid: every ITA model is connected to the puzzle model by a groupoid morphism.

This universality is satisfying but also raises a question. If the puzzle model is canonical within the groupoid, what determines which R-matrices *define* the groupoid in the first place? Which 4×4 matrices satisfy the Yang-Baxter equation? For the five-vertex models relevant to LR coefficient theory, this is a finite classification problem. We turn to it now.

4.2 The five-vertex classification and chirality

The R-matrices of Section 3.2 are 4×4 matrices acting on $\mathbb{C}^2 \otimes \mathbb{C}^2$, with edge spins in $\{0, 1\}$. The conservation law (total input spin equals total output spin) restricts the nonzero entries to the six "ice-type" configurations:

$$(0,0) \to (0,0), \quad (1,1) \to (1,1), \quad (0,1) \to (0,1),$$
$$(1,0) \to (1,0), \quad (0,1) \to (1,0), \quad (1,0) \to (0,1).$$

A *six-vertex model* assigns nonzero weights to all six; a *five-vertex model* forbids exactly one of the six, setting its weight to zero. The KTW puzzle model is a five-vertex model: it forbids the configuration $(1,0) \to (1,0)$, in which a spin-1 entering from the left and a spin-0 entering from below pass through the vertex unchanged.

The classification question

Among all five-vertex R-matrices with a spectral parameter u, which ones satisfy the Yang-Baxter equation (6)? Since there are six choices of which vertex to forbid, and for each choice the remaining five weights are functions of u to be determined, this is a finite (but nontrivial) classification problem.

The YBE is a system of 64 cubic equations in the vertex weights. Most of the 64 equations are trivially satisfied when one of the six vertices is forbidden. The content of the classification is in the remaining equations: they impose algebraic relations among the five nonzero weights, and these relations either have nontrivial solutions (the model is integrable) or they do not (the model is not integrable).

The classification result

The outcome of the classification is as follows.

1. For each choice of forbidden vertex, the five-vertex YBE system admits a family of solutions parameterised by the spectral parameter u and a finite number of structural constants (the values of the weights at a reference point).

2. Up to the symmetries of the problem (transposition of input and output, reversal of spin labels, rescaling of the R-matrix), the solutions organise into families labelled by which vertex is forbidden. The puzzle model (forbidding $(1,0) \to (1,0)$) is one such family.

3. There is essentially one nontrivial five-vertex solution satisfying the full spectral-parameter YBE:

$$R(u) \;=\; c(u)\big(P + \alpha u \cdot E_{22}\big), \tag{11}$$

where P is the permutation (swap) matrix, E_{22} is the matrix unit with a 1 in the $(2,2)$ position and zeros elsewhere, α is a structural constant, and $c(u)$ is a scalar normalisation. This R-matrix computes ω-dual Jacobi-Trudi determinants — the $t = 0$ specialisation of the Wheeler-Zinn-Justin Hall-Littlewood model [39].

Chirality

To state the classification precisely, we need the concept of *chirality*. Define the two "non-crossing" vertices as those where each spin passes straight through: $(0,1) \to (0,1)$ with weight a_1, and $(1,0) \to (1,0)$ with weight a_2. The two "crossing" vertices are $(0,1) \to (1,0)$ with weight b_1 and $(1,0) \to (0,1)$ with weight b_2. A vertex model is *chiral* if $b_1 \neq b_2$: the weights for left-to-right and right-to-left crossings differ. It is *label-asymmetric* if $a_1 \neq a_2$: the two non-crossing vertices have different weights.

In a five-vertex model, one of the six weights is set to zero. Forbidding one of the crossing vertices (say $b_2 = 0$, as in the KTW puzzle model) automatically introduces chirality: the two crossing types are no longer equivalent, since one is allowed and the other is forbidden. This raises the question: does chirality obstruct the Yang-Baxter equation?

The answer, perhaps surprisingly, is *no*. The R-matrix (11) is chiral (one crossing type is suppressed relative to the other, as the permutation matrix P treats the two asymmetrically once E_{22} is added) and it satisfies the full YBE. The KTW puzzle model itself is chiral and integrable. Chirality alone is not an obstruction to integrability.

What *is* an obstruction is the *combination* of chirality with label asymmetry. The complete five-vertex classification can be stated as a dichotomy:

> **Chirality principle.** A five-vertex model with spectral parameter satisfies the Yang-Baxter equation if and only if either
>
> (a) the non-crossing weights are equal: $a_1 = a_2$; or
>
> (b) the model is a degenerate (free-fermionic) specialisation of the six-vertex model.
>
> In particular, chirality combined with $a_1 \neq a_2$ is the *only* obstruction to the five-vertex YBE.

This principle was verified computationally for all six choices of forbidden vertex: of the 64 matrix-element equations comprising the YBE, exactly 4 produce nonzero residuals in the generic $a_1 \neq a_2$ case, and the residuals are proportional to $q(p-q)$, where p and q are the two non-crossing weights. The obstruction vanishes if and only if $p = q$ (condition (a)) or the model degenerates further (condition (b)).

The chirality principle has a clear physical interpretation. In a chiral model, paths through the lattice "know" whether they are turning left or right — the model distinguishes orientation. This is compatible with the YBE as long as the "straight-through" weights treat both spin values equally ($a_1 = a_2$). But if the straight paths *also* distinguish between spin values ($a_1 \neq a_2$), then the combination of orientational asymmetry (chirality) and spin asymmetry (label asymmetry) produces an overdetermined system with no solutions.

Remark 4.2. The chirality principle explains why the KTW puzzle model is integrable: although it is chiral (the forbidden vertex $(1,0) \to (1,0)$ breaks the crossing symmetry), its non-crossing weights are both equal to 1, so $a_1 = a_2$. Condition (a) is satisfied. The K-theoretic deformation of Section 3.3 preserves this equality (both non-crossing weights deform together), which is why the Grothendieck R-matrix also satisfies the YBE.

4.3 The three-level hierarchy

The five-vertex classification reveals that the space of integrable models is not a continuum but a stratified landscape. The R-matrices satisfying the full YBE form one stratum; below them sit R-matrices that satisfy a weaker form of the equation; below those, the permutation matrix P, which satisfies the YBE trivially. This stratification organises into a strict three-level hierarchy that connects the classification of this section to the categorical phase diagram of Section 5.

Level 1: Solvable (full YBE)

At the top of the hierarchy sit the *solvable* models: those whose R-matrix $R(u)$ satisfies the full spectral-parameter Yang-Baxter equation (6). The transfer matrices $T(u)$ form a commuting family, the model can be diagonalised by the Bethe ansatz, and R-matrix moves provide local bijections between configurations. The Yang-Baxter groupoid of Section 4.1 lives at this level.

The characters computed by solvable models are the classical symmetric functions and their deformations: Schur polynomials (rational five-vertex model), Grothendieck polynomials (trigonometric five-vertex), and Hall-Littlewood polynomials (higher-spin models). In each case, the structure constants are nonnegative — a property that can be traced to the existence of a partition function with positive Boltzmann weights.

Level 2: Quasi-solvable (factored YBE)

The middle level contains the *quasi-solvable* models, a term due to Buciumas and Scrimshaw [10]. Here the full YBE fails at the vertex level: one cannot freely interchange three crossing lines by local moves. But a weaker form of integrability persists. The row transfer matrices and the column transfer matrices each satisfy the YBE *independently* — the R-matrix factors into separate "row" and "column" components, each integrable on its own, but there is no single unified R-matrix governing the full model.

The characters at this level are the *key polynomials* (Demazure characters). Algebraically, the quasi-solvable level corresponds to the specialisation $q = 0$ of the Hecke algebra parameter, where the Hecke relation $(T_i - q)(T_i + 1) = 0$ degenerates to $T_i(T_i + 1) = 0$ and the Demazure operators $\pi_i = T_i + 1$ become idempotent: $\pi_i^2 = \pi_i$. The braid group symmetry of the solvable level degenerates to the *cactus group* J_n — a group generated by interval reversals $s_{[i,j]}$ satisfying the relations of Henriques and Kamnitzer [23] but *not* the braid relations.

Level 3: Combinatorial (trivial YBE)

At the base of the hierarchy, the R-matrix is simply the permutation matrix P: the swap map on $\mathbb{C}^2 \otimes \mathbb{C}^2$. The YBE is satisfied trivially ($P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12}$ reduces to the symmetric group relation $s_i s_j s_i = s_j s_i s_j$), but there is no spectral parameter and no interesting commuting family. The "transfer matrices" are combinatorial objects — crystal operators in the sense of Kashiwara [27] — and the characters are monomials, the building blocks from which all higher-level characters are assembled.

The hierarchy is strict

It is natural to ask whether the intermediate level is genuinely intermediate, or merely a degeneration of the solvable level that retains all of its structure. The answer is that the hierarchy is *strict*: the quasi-solvable level has less structure than the solvable level, and the difference is not an artifact of the specialisation $q = 0$.

The sharpest evidence comes from the *Zinn-Justin tetrahedron equation* (ZTE), a cubic coherence condition on the R-matrix that generalises the YBE from two dimensions to three. At the solvable level, the ZTE holds (it follows from the full parametric YBE by a standard

argument involving auxiliary spaces). At the quasi-solvable level, the ZTE *fails*: a direct computation at rank 3 shows that the residual is proportional to q^2 and transforms as an antisymmetric 2-form on the space of spectral parameters.

This failure has a precise interpretation: it measures the *curvature* of a connection on a vector bundle over the higher Bruhat order. At $q = 0$, the curvature is nonzero, meaning that parallel transport around a closed loop in the space of row orderings does not return to the identity. The holonomy of this connection is the obstruction to promoting column-level integrability to full vertex-level integrability. The quasi-solvable level is a flat $q = 0$ limit in one direction (each column is integrable) but *curved* in the transverse direction (the columns do not interleave coherently).

A structural summary

The three levels, together with their algebraic and categorical signatures, are summarised in the following table:

Level	**YBE status**	**Algebra**	**Characters**
Solvable	Full spectral-parameter YBE	H_q (generic q)	Schur, Grothendieck, HL
Quasi-solvable	Factored (column-level) YBE	H_0 ($q = 0$ Hecke)	Key / Demazure
Combinatorial	Trivial ($R = P$)	$\mathbb{C}[S_n]$	Monomials

The rightmost column reveals a striking alignment: the integrability level of the R-matrix controls which characters the model computes. Solvable models compute symmetric functions (Schur and their deformations); quasi-solvable models compute nonsymmetric but "partially symmetric" characters (key polynomials); combinatorial models compute the fully nonsymmetric monomials. The level of integrability is the level of symmetry.

Preview of Section 5

The three levels of the hierarchy correspond to three types of monoidal categories:

1. **Braided monoidal** (solvable): the R-matrix provides a braiding — a natural isomorphism $V \otimes W \xrightarrow{\sim} W \otimes V$ satisfying the braid relation. The symmetry group is the braid group B_n.

2. **Coboundary monoidal** (quasi-solvable): the braiding degenerates to a *commutor* — a natural isomorphism that satisfies the weaker cactus group relations but not the braid relation. The symmetry group is the cactus group J_n.

3. **Symmetric monoidal** (combinatorial): the commutor is the trivial swap $\tau\colon V \otimes W \to W \otimes V$, and $\tau^2 = \mathrm{id}$. The symmetry group is the symmetric group S_n.

The Hecke algebra $H_q(S_n)$ simultaneously hosts all three structures: at generic q, the Hecke generators provide a braiding; at $q = 0$, the braiding dies but the coboundary commutor (constructed from Drinfeld's unitarised R-matrix) survives; at $q = 1$ (equivalently, passing to the group algebra $\mathbb{C}[S_n]$), only the symmetric swap remains. The parameter q governs the transition between levels, and the Hecke algebra is the *transition algebra* of the hierarchy.

Making this precise — identifying what is forgotten at each transition, and proving that the forgetting is irreversible — is the content of Section 5.

5 The Categorical Phase Diagram

The three-level hierarchy of Section 4.3 — solvable, quasi-solvable, combinatorial — is a classification of R-matrices by how much of the Yang-Baxter equation they satisfy. This section reveals what the hierarchy *is*: three types of monoidal category, three symmetry groups, three character theories, governed by a single transition algebra. The passage between levels is not a loss of structure but a forgetting of structure — precise, irreversible, and mathematically natural. What emerges is a phase diagram in which the multiplicity of combinatorial models for LR coefficients is not an accident of history but a necessary consequence of the layered categorical architecture.

5.1 Three types of monoidal categories

The three levels of integrability correspond to three ways of equipping a monoidal category with a notion of "swapping" two objects. Each carries a different symmetry group, and the inclusions between them are strict.

Symmetric monoidal categories

The simplest structure is a *symmetric monoidal category*: a monoidal category equipped with a natural isomorphism $\sigma_{V,W} \colon V \otimes W \xrightarrow{\sim} W \otimes V$ satisfying $\sigma_{W,V} \circ \sigma_{V,W} = \mathrm{id}_{V\otimes W}$. The swap squares to the identity. The prototypical example is Vect, the category of finite-dimensional vector spaces with the usual transposition $v \otimes w \mapsto w \otimes v$. The symmetry group governing n-fold tensor products is the symmetric group S_n: each permutation $\pi \in S_n$ acts on $V^{\otimes n}$ by permuting tensor factors, and the relation $\sigma^2 = \mathrm{id}$ ensures that the action factors through S_n.

Braided monoidal categories

A *braided monoidal category* is a monoidal category with a natural isomorphism $\beta_{V,W} \colon V \otimes W \xrightarrow{\sim} W \otimes V$ satisfying the *hexagon axioms*, which encode the Yang-Baxter equation categorically:

$$(\beta_{V,W} \otimes \mathrm{id}_U) \circ (\mathrm{id}_V \otimes \beta_{U,W}^{-1})^{-1} \circ (\beta_{U,V} \otimes \mathrm{id}_W) \;=\; (\mathrm{id}_W \otimes \beta_{U,V}) \circ (\beta_{U,W} \otimes \mathrm{id}_V)^{-1} \circ \cdots \tag{12}$$

but *not* the involution $\beta^2 = \mathrm{id}$. More concretely: on a triple tensor product $U \otimes V \otimes W$, writing β_{12} for the braiding applied to the first two factors and β_{23} for the braiding on the last two, the hexagon axioms reduce to

$$\beta_{12} \circ \beta_{23} \circ \beta_{12} \;=\; \beta_{23} \circ \beta_{12} \circ \beta_{23}, \tag{13}$$

the braid relation. The symmetry group is the braid group B_n, whose generators $\sigma_1, \ldots, \sigma_{n-1}$ satisfy (13) but are *not* involutions. The central example is $\mathrm{Rep}(U_q(\mathfrak{g}))$ for generic q: the braiding $\beta_{V,W} = P \circ R_{V,W}$ is constructed from Drinfeld's universal R-matrix, and the fact that R satisfies the parametric YBE of Section 3.2 is precisely the braid relation (13) in categorical language.

Coboundary monoidal categories

Between symmetric and braided lies a structure with no classical analogue in physics: the *coboundary monoidal category*, introduced by Drinfeld and developed in the representation-theoretic context by Henriques and Kamnitzer [23]. A coboundary category is a monoidal category with a natural isomorphism $\sigma_{V,W} \colon V \otimes W \xrightarrow{\sim} W \otimes V$—called the *commutor*—satisfying three axioms:

(C1) *Involution:* $\sigma_{W,V} \circ \sigma_{V,W} = \mathrm{id}$. (Like a symmetric category, but unlike a braided one.)

(C2) *Disjoint commutativity:* if the intervals $[i,j]$ and $[k,l]$ are disjoint subsets of $\{1,\dots,n\}$, then the commutors $\sigma_{[i,j]}$ and $\sigma_{[k,l]}$ commute.

(C3) *Nesting:* if $[k,l] \subseteq [i,j]$, then $\sigma_{[i,j]} \circ \sigma_{[k,l]} \circ \sigma_{[i,j]} = \sigma_{[\sigma(k),\sigma(l)]}$, where σ denotes the reversal of $[i,j]$.

These are the defining relations of the *cactus group* J_n of Devadoss and Henriques-Kamnitzer [23]. The cactus group is generated by elements $s_{[i,j]}$ for $1 \le i < j \le n$, subject to (C1)–(C3). It is an infinite group (unlike S_n), and its structure is genuinely distinct from both the braid group and the symmetric group: J_n admits a surjection onto S_n (map each interval reversal $s_{[i,j]}$ to the corresponding permutation), but this surjection does *not* split.

Remark 5.1 (Non-splitting of J_3). For $n = 3$, the surjection $J_3 \twoheadrightarrow S_3$ does not admit a group-theoretic section. One can verify this directly: J_3 is generated by $s_{[1,2]}$, $s_{[2,3]}$, and $s_{[1,3]}$, with $s_{[1,2]}^2 = s_{[2,3]}^2 = s_{[1,3]}^2 = 1$ and the nesting relation $s_{[1,3]}s_{[1,2]}s_{[1,3]} = s_{[2,3]}$. Any section $S_3 \to J_3$ would need to send the transpositions (12) and (23) to elements $a, b \in J_3$ satisfying $a^2 = b^2 = 1$ and $aba = bab$ (the braid relation modulo $\sigma^2 = 1$). But in J_3, the products $s_{[1,2]}s_{[2,3]}s_{[1,2]}$ and $s_{[2,3]}s_{[1,2]}s_{[2,3]}$ are *not* equal: the nesting relation (C3) is not the braid relation. No choice of lifts satisfies all the S_3 relations simultaneously. The cactus group is not a semidirect product; the coboundary level does not retract onto the symmetric level.

The central example of a coboundary monoidal category is $\mathrm{Crys}(\mathfrak{g})$, the category of Kashiwara crystal bases [27]. The Henriques-Kamnitzer crystal commutor $\sigma^{\mathrm{crys}}_{V,W}$ is constructed from the Schützenberger involution on highest-weight crystals and extended to tensor products via a recursive procedure involving connected components of the tensor product crystal graph.

The hierarchy is strict

The three types form a chain of strict inclusions:

$$\text{symmetric} \subsetneq \text{coboundary} \subsetneq \text{braided}.$$

Every symmetric category is coboundary (the swap σ satisfies the cactus relations trivially, since $\sigma^2 = 1$ and the nesting relation collapses to a symmetric group identity). Every coboundary category can be *embedded into* a braided one via the Kamnitzer-Tingley unitarised R-matrix construction [26] (see Section 5.3 below). But neither inclusion reverses: not every braided category admits a coboundary structure (the braiding must be unitarisable), and not every coboundary category is symmetric (the non-splitting of $J_3 \twoheadrightarrow S_3$ shows the coboundary axioms are strictly weaker than the symmetric ones).

The three monoidal types correspond precisely to the three integrability levels of Section 4.3:

Monoidal type	**Swap structure**	**Symmetry group**	**Integrability**
Braided	$\beta^2 \neq \mathrm{id}$, braid relation	B_n (braid)	Solvable
Coboundary	$\sigma^2 = \mathrm{id}$, cactus relations	J_n (cactus)	Quasi-solvable
Symmetric	$\sigma^2 = \mathrm{id}$, braid = trivial	S_n (symmetric)	Combinatorial

This alignment is the first axis of the phase diagram. We now identify what is forgotten at each transition.

5.2 The Multiplicity Bundle Theorem

The passage from coboundary to symmetric is mediated by a single geometric object: the *multiplicity bundle*, whose fibre at each isotypic component of a tensor product is the Specht module S_λ. The coboundary commutor acts nontrivially on these fibres. The crystal functor kills them. This is the content of the central theorem.

Setup

Let $V = \mathbb{C}^n$ be the fundamental representation of $GL_n(\mathbb{C})$, and let $k \geq 2$. The Schur-Weyl decomposition

$$V^{\otimes k} \cong \bigoplus_{\substack{\lambda \vdash k \\ \ell(\lambda) \leq n}} V_\lambda \otimes S_\lambda \tag{14}$$

decomposes $V^{\otimes k}$ as a $(GL_n \times S_k)$-bimodule, with V_λ the irreducible GL_n-module and S_λ the Specht module for S_k. We think of (14) as a vector bundle over the set of partitions λ: the "base" fibre is V_λ (the physical, or representation-theoretic, factor) and the "multiplicity" fibre is S_λ (the combinatorial factor that records how many copies of V_λ appear in the tensor product).

For an interval $[i, j] \subseteq \{1, \ldots, k\}$, define the *interval reversal* $w_0^{[i,j]} \in S_k$ to be the longest element of the parabolic subgroup $S_{\{i,i+1,\ldots,j\}}$: the permutation that reverses the positions $i, i+1, \ldots, j$ and fixes all others. In particular, $w_0^{[1,k]}$ is the longest element of S_k. The interval reversal acts on $V^{\otimes k}$ by permuting tensor factors: $w_0^{[i,j]}$ sends $v_1 \otimes \cdots \otimes v_k$ to $v_1 \otimes \cdots \otimes v_{i-1} \otimes v_j \otimes v_{j-1} \otimes \cdots \otimes v_i \otimes v_{j+1} \otimes \cdots \otimes v_k$.

Theorem 5.2 (Multiplicity Bundle)**.** *The coboundary commutor $\sigma_{[i,j]}$ on $V^{\otimes k}$ decomposes under (14) as*

$$\sigma_{[i,j]} = \bigoplus_\lambda \big(\mathrm{id}_{V_\lambda} \otimes \rho_{S_\lambda}(w_0^{[i,j]})\big), \tag{15}$$

where ρ_{S_λ} denotes the Specht module representation of S_k.

Proof sketch. The commutor $\sigma_{[i,j]}$ is, by definition, the interval reversal $w_0^{[i,j]}$ acting on $V^{\otimes k}$ by permutation of tensor factors. This action commutes with the diagonal GL_n-action (since permuting tensor factors commutes with applying the same group element to each factor). By Schur's lemma, any GL_n-endomorphism of $V^{\otimes k}$ decomposes as $\bigoplus_\lambda(\mathrm{id}_{V_\lambda} \otimes \phi_\lambda)$ for some $\phi_\lambda \in \mathrm{End}(S_\lambda)$. Since the commutor is precisely the action of $w_0^{[i,j]} \in S_k$, and the Schur-Weyl decomposition identifies the S_k-action with $\mathrm{id}_{V_\lambda} \otimes \rho_{S_\lambda}$, we obtain $\phi_\lambda = \rho_{S_\lambda}(w_0^{[i,j]})$. □

The theorem is essentially a corollary of Schur-Weyl duality. Its content lies not in the proof — which is short — but in the *interpretation*: the commutor lives entirely in the multiplicity fibres S_λ. It acts as the identity on the representation-theoretic factor V_λ and as the longest-element involution $\rho(w_0)$ on the Specht module.

Crystal invisibility

The crystal functor $\mathcal{F}\colon \mathrm{Rep}(GL_n) \to \mathrm{Crys}(\mathfrak{sl}_n)$ sends a representation to its crystal basis, retaining the combinatorial skeleton (weights, raising and lowering operators) while forgetting the linear algebra. On tensor products of the fundamental representation, the crystal functor has a decisive effect on the multiplicity bundle.

Proposition 5.3 (Crystal multiplicity-freeness)**.** *The tensor product crystal $B(\omega_1)^{\otimes k}$, where $B(\omega_1)$ is the crystal of the fundamental representation of $\mathfrak{sl}_n$ with $n \geq k$, is* multiplicity-free*: each connected component is isomorphic to a highest-weight crystal $B(\lambda)$ for a distinct partition $\lambda \vdash k$.*

In the language of the multiplicity bundle: when $n \geq k$, each fibre S_λ is one-dimensional as a crystal (it contributes exactly one copy of $B(\lambda)$). The crystal functor replaces the potentially large Specht module S_λ with a single point. Every endomorphism of a one-dimensional set is the identity. Therefore:

Corollary 5.4 (Crystal commutor is trivial)**.** *The crystal commutor $\mathcal{F}(\sigma_{[i,j]})$ is the identity on $B(\omega_1)^{\otimes k}$ for $n \geq k$. In particular, the crystal category of fundamental representations is symmetric monoidal.*

This is the mechanism by which the coboundary structure becomes invisible at the crystal level. The commutor $\sigma_{[i,j]}$ acts nontrivially — the longest element w_0 is not the identity in any Specht module of dimension greater than one — but its action is confined to fibres that the crystal functor collapses. The Specht module is the vessel that carries the coboundary structure; the crystal functor drains it.

Remark 5.5. The restriction $n \geq k$ ensures multiplicity-freeness and hence triviality of the crystal commutor on fundamental tensor products. For general representations, the crystal commutor is nontrivial — it is the Henriques-Kamnitzer commutor, constructed from the Schützenberger involution. The general crystal category $\mathrm{Crys}(\mathfrak{g})$ is coboundary monoidal, not symmetric. The symmetric monoidal structure is a special feature of the fundamental representations, where multiplicity-freeness makes the coboundary commutor degenerate. This is consistent with the fact that the lattice models of Section 3 use fundamental representations (edge spins in $\{0, 1\} = B(\omega_1)$): the combinatorial level of the hierarchy lives in the multiplicity-free regime.

The two transitions

Theorem 5.2 identifies what is forgotten at each step of the hierarchy:

Braided → Coboundary: forget the spectral parameter. At generic q, the category $\mathrm{Rep}(U_q(\mathfrak{g}))$ carries both a braiding $\beta_{V,W} = P \circ R_{V,W}$ (from the universal R-matrix, depending on a spectral parameter u) and a coboundary commutor $\sigma_{V,W}$ (from the unitarised R-matrix of Kamnitzer and Tingley [26]). These are *different* structures: β satisfies the braid relation but $\beta^2 \neq \mathrm{id}$; σ is an involution satisfying the cactus relations but not the braid relation. At $q = 0$, the braiding dies — the R-matrix degenerates and the braid relation can no longer be formed — but the commutor σ survives, because its construction uses only the unitarised (eigenvalue-independent) part of the R-matrix. The passage from braided to coboundary is not a creation of new structure but an *extraction*: the coboundary commutor was always present, and the specialisation $q = 0$ strips away the braided layer to reveal it.

Coboundary → Symmetric: forget the multiplicity bundle. The commutor $\sigma_{[i,j]} = \mathrm{id}_{V_\lambda} \otimes \rho_{S_\lambda}(w_0^{[i,j]})$ acts nontrivially on S_λ whenever $\dim S_\lambda > 1$. The crystal functor kills S_λ (Proposition 5.3), collapsing the commutor to the identity. What was a rich action of the cactus group J_n on multiplicity spaces becomes the trivial action of S_n.

These two forgettings are irreversible. The spectral parameter cannot be recovered from the commutor (many braidings can produce the same commutor via unitarisation). The multiplicity bundle cannot be recovered from the crystal (the crystal retains only the GL_n-isotypic decomposition, not the S_k-module structure within each isotypic component). Each transition is a genuine loss of information.

5.3 The Hecke Transition Algebra

The three monoidal types are governed by a single algebraic object: the Hecke algebra $H_q(S_k)$ and its degenerations. The algebra chain

$$H_q(S_k) \xrightarrow{q \to 0} H_0(S_k) \xrightarrow{\text{semisimplify}} \mathbb{C}[S_k] \tag{16}$$

mirrors the categorical hierarchy step by step. The Hecke algebra is not a fourth level of integrability; it is the *algebra of the transition itself.*

Level	Category type	Symmetry group	Fiber functor	YBE status	Characters
Solvable	Braided monoidal	B_k (braid group)	Rep $\to$ Vect	Full YBE	Macdonald P_λ
Quasi-solvable	Coboundary monoidal	J_k (cactus group)	Rep $\to$ Crys	Column YBE	Key κ_α
Combinatorial	Symmetric monoidal	S_k (symmetric group)	Crys $\to$ Set	None (trivial)	Schur s_λ

Generic q: the braided level

At generic q, the Hecke algebra $H_q(S_k)$ is the quotient of the group algebra of the braid group B_k by the Hecke relation $(T_i - q)(T_i + 1) = 0$. The generators $T_1, \ldots, T_{k-1}$ satisfy the braid relation

$$T_i T_{i+1} T_i \;=\; T_{i+1} T_i T_{i+1}$$

and the quadratic relation $T_i^2 = (q-1)T_i + q$. In the lattice model context of Section 3, the Hecke generator T_i is the *R-matrix* $\check{R}_i(u)$ acting on the i-th and $(i+1)$-th tensor factors: the braid relation is the Yang-Baxter equation, and the quadratic relation encodes the two nonzero eigenvalues q and -1 of the R-matrix. The existence of two distinct eigenvalues is what makes the braiding non-involutive ($\beta^2 \neq \mathrm{id}$) and gives the braid group its infinite order.

The characters of H_q-modules, traced through the natural representation on polynomial rings, yield the Macdonald polynomials $P_\lambda(x; q, t)$ (at the most general level) and their specialisations: Hall-Littlewood polynomials at $q = 0$, Schur polynomials at $q = t$, and so on. The solvable lattice models of Section 3 compute these characters as partition functions.

$q = 0$: the coboundary level

At $q = 0$, the Hecke relation becomes $T_i(T_i + 1) = 0$, or equivalently $T_i^2 = -T_i$. The Demazure operators $\pi_i = T_i + 1$ satisfy

$$\pi_i^2 = \pi_i, \tag{17}$$

the idempotent relation. This is the algebraic signature of the coboundary level: the R-matrix has collapsed to a *projector*, and the two eigenvalues q and -1 have merged to 0 and -1. The braid relation among the T_i persists formally, but it now coexists with idempotency, and the combination produces the cactus group rather than the braid group.

More precisely, define the cactus generators

$$\sigma_{[a,b]} \;=\; \pi_{w_0^{[a,b]}} \;=\; (T_{w_0^{[a,b]}} + 1)(T_{w_0^{[a,b]}} + 1)\cdots \tag{18}$$

as the Demazure operator indexed by the longest element $w_0^{[a,b]}$ of the parabolic subgroup $S_{\{a,\ldots,b\}}$. These operators satisfy the cactus relations (C1)–(C3): involution ($\sigma_{[a,b]}^2 = \mathrm{id}$ on suitable quotients), disjoint commutativity, and nesting. The verification is a direct computation in H_0, confirmed numerically for $k \leq 5$.

Norton [32] showed that $H_0(S_k)$ has 2^{k-1} simple modules, indexed by compositions $\alpha \models k$ rather than partitions $\lambda \vdash k$. The characters of these simple modules are the *Demazure atoms* A_α: the nonsymmetric polynomials from which key polynomials are built. The key polynomial κ_α is the character of the corresponding projective indecomposable H_0-module, and the decomposition

$$s_\lambda \;=\; \sum_{\alpha \sim \lambda} A_\alpha \tag{19}$$

— where the sum runs over compositions α that rearrange to λ — is the character-level shadow of the semisimplification $H_0 \to \mathbb{C}[S_k]$.

Semisimplification: the symmetric level

The group algebra $\mathbb{C}[S_k]$ is the semisimplification of $H_0(S_k)$: it identifies all compositions rearranging to the same partition. At the character level, this identification is (19): the Schur polynomial is the sum of Demazure atoms over compositions in the same S_k-orbit. At the module level, the 2^{k-1} simple H_0-modules (indexed by compositions) collapse to the $p(k)$ simple $\mathbb{C}[S_k]$-modules (indexed by partitions), where $p(k)$ is the number of partitions of k.

This collapse is the algebraic avatar of the crystal functor. Kashiwara's crystal basis theory retains only the partition-indexed data (the highest-weight structure), discarding the finer composition-indexed data (the Demazure filtration). The equation $s_\lambda = \sum_\alpha A_\alpha$ records the cost: the internal structure of each key polynomial, its decomposition into atoms indexed by distinct compositions, is forgotten when one passes to the Schur level.

The Kamnitzer-Tingley subtlety

There is a crucial point that prevents the hierarchy from being merely a story about specialising q. Kamnitzer and Tingley [26] showed that the coboundary commutor exists at *all* values of q, not only at $q = 0$. Their construction uses Drinfeld's *unitarised R-matrix*: starting from the universal R-matrix $\mathcal{R}$ of $U_q(\mathfrak{g})$, one extracts an involutive commutor $\sigma_{V,W} = P \circ \mathcal{R}^{\text{unit}}_{V,W}$ that satisfies the cactus relations for all q.

This means that the category $\text{Rep}(U_q(\mathfrak{g}))$ is *simultaneously* braided monoidal (via the R-matrix) and coboundary monoidal (via the unitarised R-matrix) for all generic q. The two structures coexist. At $q = 0$, the braiding dies — the R-matrix degenerates, the two eigenvalues of T_i collide, and the braid relation loses its force — but the coboundary commutor survives intact. Debraiding is not a creation but an *extraction*: one peels away the braided layer to reveal the coboundary substrate that was always there, like removing a polarising filter to see the light that was passing through it all along.

Remark 5.6. This subtlety has a direct consequence for the phase diagram: the coboundary column is not a "degenerate" version of the braided column. Models at the coboundary level compute different characters (key polynomials instead of Schur polynomials) and have different combinatorial structure (column-level rather than full YBE). The coboundary structure is *extracted* from the braiding, not *created* by degeneration. The two levels are related by a forgetful functor, not by a limiting process.

Remark 5.7 (Curvature unification). The two obstruction results in this section and Section 5.4 are instances of a single geometric object.

Braid obstruction [1]: In $H_q(S_k)$, the Demazure operators $\pi_i = T_i + 1$ satisfy

$$\pi_i\pi_{i+1}\pi_i - \pi_{i+1}\pi_i\pi_{i+1} \;=\; q(T_i - T_{i+1}).$$

This is antisymmetric in $i \leftrightarrow i+1$ and linear in q.

ZTE curvature (April 2026 computation): The Zamolodchikov tetrahedron equation fails for the puzzle R-matrices at rank ≥ 3. The failure is measured by a curvature proportional to q^2, antisymmetric on the higher Bruhat order $B(n, 2)$. At $q = 0$, the curvature vanishes.

Both are the *curvature of the Demazure connection on the higher Bruhat order*. The braid obstruction is the infinitesimal curvature (first order in q); the ZTE failure is the finite curvature (second order, from composing two first-order obstructions). The factor of q^2 in the ZTE is the product of two factors of q from two successive braid obstructions, which is consistent with the holonomy formula for a connection with curvature linear in q.

The geometric content: $q = 0$ *is the flat locus of the Demazure connection.* The coboundary monoidal structure lives at the flat point. The braided structure lives in the curved region. The passage from braided to coboundary is not a degeneration but a *flattening*: moving from a

curved connection to a flat one, where path-independence of the puzzle assembly (equivalently, word-independence of π_w) is restored.

5.4 The Cactus Midpoint

The algebraic transition between levels has a quantitative counterpart, recently revealed by Tikhonov's analysis of random sorting networks [38]. Consider the interpolating operator in the Hecke algebra:

$$R_i(p) \;=\; p\,T_i + (1-p) \;\in\; H_q(S_k). \tag{20}$$

At $p = 1$, this is the Hecke generator T_i; at $p = 0$, the identity. The parameter p interpolates between "doing nothing" and "applying the full R-matrix braiding."

Rewriting in terms of the Demazure operator $\pi_i = T_i + 1$:

$$R_i(p) = p(\pi_i - 1) + (1-p) = p\,\pi_i + (1-2p).$$

At $p = 1/2$, the constant term vanishes:

$$R_i(1/2) \;=\; \tfrac{1}{2}\,\pi_i. \tag{21}$$

The midpoint of the interpolation is a scalar multiple of the Demazure projector.

Proposition 5.8 (Cactus at the midpoint)**.** *Specialise to $q = 0$, so that $\pi_i^2 = \pi_i$. Let $w_0 \in S_k$ have a reduced word $w_0 = s_{i_1} \cdots s_{i_\ell}$ with $\ell = \ell(w_0) = \binom{k}{2}$. Then*

$$\prod_{j=1}^{\ell} R_{i_j}(1/2) \;=\; 2^{-\ell(w_0)} \cdot \sigma_{[1,k]}, \tag{22}$$

where $\sigma_{[1,k]} = \pi_{w_0}$ is the cactus generator corresponding to the full reversal. More generally, for any interval $[a,b]$, the product over a reduced word for $w_0^{[a,b]}$ at $p = 1/2$ recovers the cactus generator $\sigma_{[a,b]}$ up to the scalar $2^{-\ell(w_0^{[a,b]})}$.

Proof sketch. At $q = 0$, each factor $R_{i_j}(1/2) = \frac{1}{2}\pi_{i_j}$ is a scaled idempotent. The product $\prod \frac{1}{2}\pi_{i_j} = 2^{-\ell} \prod \pi_{i_j}$. By a standard property of the 0-Hecke monoid, the product $\pi_{i_1} \cdots \pi_{i_\ell}$ over any reduced word for w_0 equals π_{w_0}, which is the cactus generator $\sigma_{[1,k]}$. The key point is that idempotency ($\pi_i^2 = \pi_i$) makes the product *independent of the choice of reduced word*: different reduced decompositions of w_0 give the same operator π_{w_0}, because any two reduced words are connected by braid moves, and in the 0-Hecke monoid, braid-equivalent reduced words yield equal products. □

Proposition 5.9 (Universal scaled projection)**.** *For* all *values of q, the operator $R_i(1/2)$ satisfies*

$$R_i(1/2)^2 \;=\; \frac{q+1}{2}\,R_i(1/2).$$

Thus $R_i(1/2)$ is a scaled projection onto the q-eigenspace of T_i, universally in q. However, the product over a reduced word for w_0 telescopes to the cactus generator only *at $q = 0$, where idempotency makes the product word-independent.*

The two propositions together paint a striking picture. The parameter p interpolates between identity and braiding, and the coboundary level sits at the exact midpoint:

p	**Operator**	**Level**
0	identity	trivial
1/2	cactus generator $\sigma_{[1,k]}$ (at $q = 0$)	coboundary
1	Hecke longest element T_{w_0}	braided

The coboundary structure is not at the boundary of the braided interpolation but at its *centre*.

Remark 5.10 (Proof and word-independence). Proposition 5.8 is proved in detail in our companion note [1]. The proof proceeds in two steps.

1. At $q = 0$, one establishes the *braid relation for* π_i:

$$\pi_i\pi_{i+1}\pi_i \;=\; \pi_{i+1}\pi_i\pi_{i+1} \quad (\text{in } H_0(S_k)),$$

   which follows from the braid obstruction $\pi_i\pi_{i+1}\pi_i - \pi_{i+1}\pi_i\pi_{i+1} = q(T_i - T_{i+1})$ (see Remark 5.7 and [1]) vanishing at $q = 0$.

2. Combined with idempotency $\pi_i^2 = \pi_i$ (which also holds only at $q = 0$), this and Matsumoto's theorem give word-independence of $\pi_{w_0^{[a,b]}}$. The midpoint identity $R_i(1/2) = \frac{1}{2}\pi_i$ then yields (22) by straightforward algebra.

At generic q, both idempotency and the braid relation fail, so the product $\prod R_{i_j}(1/2)$ is word-dependent and does not yield a well-defined cactus generator. The cactus midpoint is *special to* $q = 0$ for provably algebraic reasons, not as a limiting accident.

Remark 5.11 (Galashin gap). Galashin [17] introduced the *Yang-Baxter basis* of the Hecke algebra $H_q(S_k)$, whose elements are products of the operators $R_i(p) = pT_i + (1-p)$ (our interpolating operators) indexed by reduced words. He showed that these basis elements compute partition functions of the stochastic six-vertex model. As of 2026, the paper has accumulated 26 citations. *None* of those 26 papers connect the Yang-Baxter basis to cactus groups, coboundary monoidal categories, or the coboundary structure of the $q = 0$ Hecke algebra.

Proposition 5.8 fills this gap precisely: it identifies the $p = 1/2$ specialisation of the Yang-Baxter basis element $\prod R_{i_j}(p)$ (for a reduced word of $w_0^{[a,b]}$) as a scalar multiple of the cactus generator $\sigma_{[a,b]}$, and it proves that this identification works only at $q = 0$. The categorical phase diagram thus has a precise algebraic location in the Yang-Baxter basis interpolation: coboundary = midpoint, braided = endpoint, and the passage between them is the Demazure connection curvature of Remark 5.7.

More specifically: Galashin's operators $R_k(p) = pT_k + (1 - p)$ (his eq. 2.4, originating in Lascoux-Leclerc-Thibon [18]) are *identical* to our interpolating operators. Galashin's special values are the physical crossing probabilities $p_{i,j} = (y_j - x_i)/(y_j - qx_i)$ (the integrable S6V weights), the Kazhdan-Lusztig limit $p = 1/q$, and pipe dream substitutions. The value $p = 1/2$ does not appear. His central result — the flip theorem (Theorem 1.1) — is a distributional symmetry of the S6V model under 180-degree rotation with rapidity reversal. At $q = 0$ and $p = 1/2$, this flip theorem restricts to a statement about the coboundary commutor acting on crystal bases, connecting Galashin's probabilistic symmetry to Halacheva-Kamnitzer-Rybnikov-Weekes's cactus group action through the common language of $H_0(S_k)$.

Connection to the Tikhonov permuton

Tikhonov [38] studies the limiting permuton of random sorting networks in the (p, q) parameter space, where p is the interpolation parameter and q is the Hecke deformation. The main result is that the limiting shape depends on (p, q) only through the invariant

$$\kappa \;=\; \frac{1 - qp}{1 - p}, \tag{23}$$

and the Möbius formula

$$p' \;=\; \frac{p(1 - q)}{1 - qp} \tag{24}$$

relates different (p, q) values producing the same permuton. At the cactus midpoint ($p = 1/2$, $q = 0$), the invariant takes the value $\kappa = 2$.

Remark 5.12 (A coincidence worth watching). The Möbius formula (24) is an element of PGL(2) — the same group that acts on spectral parameters in integrable systems. In the lattice model context, the spectral parameter u of the R-matrix transforms under the action of the Yangian, and the Yang-Baxter equation is covariant under Möbius transformations of u. The appearance of PGL(2) in Tikhonov's formula *and* in the spectral parameter theory may not be a coincidence: if the two PGL(2) actions are related, the categorical phase diagram would acquire a quantitative prediction linking the interpolation parameter p to the spectral parameter u. We flag this as an open question.

Remark 5.13 (Rank hierarchy of the cactus midpoint). The three categorical levels can be *quantified* by the rank of three operators in the left regular representation of $H_q(S_n)$ (dimension $n!$):

Operator	$q = 0$ **rank**	**Generic** q **rank**	**Category**
q-symmetriser $\sum_w T_w$	1	1	Symmetric
Cactus midpoint $\prod(T_{i_j} + 1)$	1	$n!/2^{\lfloor n/2 \rfloor}$	Coboundary
T_{w_0}	1	$n!$ (invertible)	Braided

The cactus midpoint element $\Pi = \prod_{j=1}^{\ell(w_0)}(T_{i_j} + 1)$, taken over any reduced word $s_{i_1} \cdots s_{i_m}$ for w_0, has the following representation-theoretic structure at generic q: the sign representation $V_{(1^n)}$ is killed (since $T_i + 1$ annihilates the (-1)-eigenspace), the trivial representation survives with rank 1, and the total weighted rank $\sum_\lambda r_\lambda \dim V_\lambda$ equals $n!/2^{\lfloor n/2 \rfloor}$. More precisely, V_λ is killed if and only if $\ell(\lambda) > (n+1)/2$: partitions with too many parts are annihilated because the accumulated projections onto $(+1)$-eigenspaces of adjacent transpositions leave no room for column-heavy tableaux. Verified explicitly for $n = 2$ through 7:

$$1,\ 3,\ 6,\ 30,\ 90,\ 630 \qquad \text{for } S_2, S_3, S_4, S_5, S_6, S_7.$$

This is OEIS sequence `A090932`. Callan observed that $a(n)$ counts the permutations of $[n]$ in which every ascent begins at an even position—a characterisation that may give a direct combinatorial description of the image of Π.

A key simplification: the rank is *q-independent*. Writing $P_i = (T_i + 1)/(q + 1)$, each P_i is idempotent at every q, so the rank of Π at generic q equals that of the corresponding product $\prod(s_{i_j} + 1)/2$ in $\mathbb{C}[S_n]$. The question thus reduces entirely to the symmetric group.

The *element* Π depends on the choice of reduced word for w_0. For most reduced words—including the staircase word—the rank equals $n!/2^{\lfloor n/2 \rfloor}$, but a small number of *anomalous* words achieve higher rank (exactly double in all observed cases: 4 of 16 words in S_4, 4 of 768 in S_5). These anomalous words over-represent the middle generator $s_{\lfloor n/2 \rfloor}$. The word-dependence is a direct consequence of the braid obstruction (Remark 5.7): since $(1+s_i)(1+s_{i+1})(1+s_i) \neq (1+s_{i+1})(1+s_i)(1+s_{i+1})$, Matsumoto's theorem does not apply, and different reduced words yield genuinely different group algebra elements. The minimal rank $n!/2^{\lfloor n/2 \rfloor}$ is the generic value. In Galashin's Yang–Baxter basis, the staircase-word operator takes the form $\Pi = 2^{\ell(w_0)} \cdot Y^{w_0}(p = 1/2)$. At $q = 0$, idempotency collapses both rank and element to word-independence. The rank thus measures *how much of the Hecke algebra survives* through each layer of the phase diagram.

Theorem 5.14 (H-invariant characterisation of the rank)**.** *Let $H = \langle s_1, s_3, s_5, \ldots \rangle \leq S_n$ be the subgroup generated by the transpositions swapping consecutive pairs of positions: $H \cong S_2^{\lfloor n/2 \rfloor}$. For each partition $\lambda \vdash n$, let r_λ denote the rank of the staircase-word midpoint element $\Pi = \prod_{j=1}^{\ell(w_0)}(1 + s_{i_j})/2$ acting on the irreducible representation V_λ of S_n. Then*

$$r_\lambda \ = \ \dim V_\lambda^H,$$

where $V_\lambda^H = \{v \in V_\lambda : h \cdot v = v \text{ for all } h \in H\}$ is the subspace of H-fixed vectors.

Remark 5.15 (Rank theorem, not functor isomorphism). Theorem 5.14 is a statement about *dimensions*: $r_\lambda = \dim V_\lambda^H$ in every irreducible representation. It is *not* the assertion that the subspaces coincide. In fact $\mathrm{im}(\Pi|_{V_\lambda}) \neq V_\lambda^H$ already for small n—the staircase product does not commute with all generators of H (see Remark 5.16 below). Two functors, equal dimensions in every irrep, different images: the question of whether a natural isomorphism between them exists is a genuinely deeper problem, posed in Section 7 (Question Q4).

The theorem is proved in [2]; a computational verification for all $\lambda \vdash n$ with $n \leq 16$ provided the empirical basis. The proof proceeds by a *bypass* route: one shows directly that every eigenvalue of Π_q on each irrep is strictly positive for $q > 0$ (see [3]), using non-negativity of the Kazhdan–Lusztig W-graph together with a Perron–Frobenius argument on an exterior power; coupled with a counting of column-ascending-free standard Young tableaux, this forces $\mathrm{rk}(\Pi_1|_{V_\lambda}) = \dim V_\lambda^H$ without having to close the Frobenius-squeeze argument block by block. The three corollaries of Remark 5.13 follow immediately:

(i) **Rank formula.** By Frobenius reciprocity, $\sum_\lambda \dim V_\lambda \cdot \dim V_\lambda^H = |S_n/H| = n!/2^{\lfloor n/2 \rfloor}$.

(ii) **Vanishing criterion.** $\dim V_\lambda^H = 0$ if and only if $\ell(\lambda) > (n+1)/2$, by a pigeonhole argument: H acts on $\lfloor n/2 \rfloor$ pairs of positions, and any H-invariant vector in the Specht module must symmetrise each pair, but column-heavy tableaux cannot accommodate that many symmetrisations.

(iii) **Hook formula.** For hook partitions $\lambda = (n-k, 1^k)$, we have $V_\lambda \cong \bigwedge^k V_{\mathrm{std}}$, and $\dim(\bigwedge^k V_{\mathrm{std}})^H = \binom{\lfloor (n-1)/2 \rfloor}{k}$.

Remark 5.16 (Two functors, same dimension). The staircase product Π does *not* commute with the generators $s_3, s_5, \ldots$ of H, so $\mathrm{im}(\Pi|_{V_\lambda}) \neq V_\lambda^H$ in general. Yet these two subspaces have the same dimension in every irreducible representation—the hallmark of a natural isomorphism between two functors $V_\lambda \mapsto \mathrm{im}(\Pi|_{V_\lambda})$ and $V_\lambda \mapsto V_\lambda^H$. Understanding this isomorphism is an important open problem (Question Q4).

Remark 5.17 (Proof architecture). The H-invariant theorem admits a proof via Frobenius reciprocity ("squeeze"): the lower bound $\mathrm{rk}(\Pi|_{V_\lambda}) \geq \dim V_\lambda^H$ follows from injectivity of Π on V_λ^H, which reduces to showing that each projection $P_j = (1 + s_{i_j})/2$ in the staircase product is injective on the transported V^H image. Three mechanisms cover different steps:

(a) *Within-block cascade* (proved): consecutive staircase steps involve adjacent transpositions, and $E_i^{(+1)} \cap E_{i+1}^{(-1)} = \{0\}$ in any S_n representation.

(b) *Odd block starts* (proved): at the first appearance of an H-generator s_{2m+1}, a contraction argument in Young's orthogonal form shows the transported V^H image avoids $\ker P_{2m+1}$. The proof uses the metric structure: $\|Qx\| \leq \|x\|$ with equality forcing a contradiction.

(c) *Even block* $k = 4$ (proved): at this step the image lies in $E_1^{(+1)} \cap E_3^{(+1)}$; adjacent disjointness applied twice (with s_1 and s_3) shows $\ker((1+s_4)(1+s_2))$ is trivial on this subspace.

Combined with the total rank formula (image basis theorem, proved in [2]), Frobenius reciprocity forces equality per irrep on the cases (a)–(c). The even-block cases $k \geq 6$ are *not* closed by a block-by-block Frobenius squeeze—but this no longer matters. The eigenvalue-positivity theorem of [3] bypasses them entirely: positivity of every eigenvalue of Π_q for $q > 0$ in every irrep, combined with an exterior-power / CA-free SYT count, yields the rank formula uniformly in λ. Thus the only role of the block-by-block argument in (a)–(c) is to give an independent, more elementary proof of the theorem in low blocks; the full theorem is established by the bypass route.

A note on how the argument found its own path. The original plan treated the H-invariant theorem as a *combinatorial* problem: close the even-block gap at $k \geq 6$ by extending cases (a)–(c) through a finer Frobenius-squeeze analysis of how staircase factors interact with the tableau combinatorics of V_λ. The proof that succeeded runs along a different axis. Eigenvalue positivity via W-graph/Kazhdan–Lusztig non-negativity is a *general* fact about products of factors $(1+T_s)$ in the Hecke algebra—it does not care that our particular product is a staircase word. The even-block gap thereby loses its status as an obstruction and becomes a standalone combinatorial question: *is there a direct tableau-level witness to the even-block cancellations?* Interesting in its own right, no longer required for the theorem.

Remark 5.18 (The pair subgroup as coboundary shadow). The subgroup $H = S_2^{\lfloor n/2 \rfloor}$ has a natural interpretation in the categorical phase diagram: it is the *coboundary shadow* of S_n. The braided-to-coboundary transition forgets the spectral parameter, collapsing the braid group B_n to the cactus group J_n. At the level of the symmetric group, this collapse is measured by H: the H-fixed subspace V_λ^H is exactly the part of V_λ that "survives" the transition from braided to coboundary monoidal structure. We are not aware of prior work studying the invariant theory of this specific subgroup.

Remark 5.19 (Hecke rank constancy and eigenvalue factorization). The staircase product Π lifts to the Iwahori–Hecke algebra $H_q(S_n)$ as $\Pi_q = \prod_j (1 + T_{i_j})$, where $T_i^2 = (q-1)T_i + q$. Computational evidence (verified for $n \leq 8$) suggests that $\operatorname{rank}(\Pi_q|_{V_\lambda(q)}) = \dim(V_\lambda^H)$ for all $q \notin \{0, -1\}$, i.e., the rank is *constant* in the Hecke parameter.

More precisely, for each $\lambda \vdash n$ with $\dim(V_\lambda^H) > 0$, the nonzero eigenvalue polynomials of Π_q on V_λ appear to factor as

$$q^{n(\lambda)} \cdot (1+q)^{\lfloor n/2 \rfloor} \cdot p_\lambda(q),$$

where $n(\lambda) = \sum_{i \geq 1} (i-1)\lambda_i$ and $p_\lambda(q)$ is a *palindromic* polynomial with positive integer coefficients satisfying $p_\lambda(q) > 0$ for all $q > 0$ (verified for $n \leq 6$). The palindromic structure allows a reduction via $u = q + 1/q$: positivity for $q > 0$ is equivalent to $h_\lambda(u) > 0$ for $u \geq 2$, where $p_\lambda(q) = q^{\deg(p)/2} h_\lambda(u)$. If proved, this would establish the H-invariant theorem (Theorem 5.14) for all n via the Frobenius squeeze argument.

5.5 Three Specializations of Π_q

The staircase product is most naturally viewed as a single element of the Iwahori–Hecke algebra $H_n(q)$:

$$\Pi_q \;:=\; (1+T_1)(1+T_2)\cdots(1+T_{n-1}),$$

where the generators T_s satisfy the quadratic relation $(T_s - q)(T_s + 1) = 0$. Specialising the parameter q gives three qualitatively distinct pictures of the *same* element, each belonging to a different world of the phase diagram.

- **At $q = 0$ (nil-Coxeter algebra).** The relation becomes $T_s^2 = -T_s$, so the T_s are (up to sign) idempotents, and $\Pi_0 = (1+T_1)\cdots(1+T_{n-1})$ is an element of the nil-Coxeter algebra. By Fomin–Stanley's construction, this is precisely the operator whose action on the polynomial ring computes the Schubert polynomial $\mathfrak{S}_{w_0}$ of the longest element. The rank behaviour here belongs to the coboundary / nil-Coxeter world—the home of the Puzzle and Lattice seed paths at $q = 0$.

- **At $q = 1$ (group algebra $\mathbb{C}[S_n]$).** The quadratic relation degenerates to $(T_s - 1)(T_s + 1) = 0$, and T_s becomes the transposition s; the element $\Pi_1 = \prod_s (1+s)$ is the classical staircase product. Its rank on the irreducible V_λ is $\dim V_\lambda^H$ by Theorem 5.14, where $H = \langle s_1, s_3, s_5, \ldots \rangle$. Here we are in the purely symmetric-group world, and V_λ^H is a Frobenius-stable subspace of each irrep.

- **At generic $q > 0$ (Hecke algebra).** The rank of $\Pi_q|_{V_\lambda(q)}$ is *constant* in q on the positive real axis and equals $\dim V_\lambda^H$. This is proved in [3]: every eigenvalue of Π_q on any irrep is strictly positive for $q > 0$, by a Perron–Frobenius argument applied to an exterior power using the non-negativity of the W-graph matrix entries (Kazhdan–Lusztig 1979). The rank is therefore lower-semicontinuous in q on $(0, \infty)$, and the rank established at $q = 1$ propagates along the whole ray.

The picture is this: *rank constancy* holds across $q > 0$—a purely Hecke-algebraic fact—but *the meaning of the rank shifts.* At $q = 0$ one is computing Schubert polynomials; at $q = 1$ one is computing H-invariant dimensions; the theorem is that these answers agree. A single numerical invariant is shared between two a priori unrelated worlds. This is the Fomin–Stanley three-worlds picture in [2], and it is what MathOverflow question Q146931 was reaching for.

Remark 5.20 (One element, three faces). The staircase product is a single element of the Hecke algebra with three faces. At $q = 0$ it is a Schubert calculator; at $q = 1$ it is an invariant-theoretic projector; in between, it is neither and both—its rank indifferent to the parameter, its interpretation sensitive to it.

Remark 5.21 (Eigenvalue Rosetta Stone). The three faces leave a trace at the level of spectra. For each $\lambda \vdash n$, computation through $n = 8$ shows that every eigenvalue of Π_q on V_λ factors as

$$\text{eigenvalue} \;=\; q^{n(\lambda)} \cdot (1+q)^{\lfloor n/2 \rfloor} \cdot p_\lambda(q),$$

where $n(\lambda) = \sum_i (i-1)\lambda_i$ is the standard partition statistic, and $p_\lambda(q)$ is a palindromic polynomial with non-negative integer coefficients and all roots in the closed left half-plane $\mathrm{Re}(z) \le 0$. We read this as a three-factor *Rosetta Stone* tying together the seed paths that run through this paper:

(i) $q^{n(\lambda)}$ is the *nil-Hecke / Schubert* contribution; $n(\lambda)$ measures distance from triviality in dominance order, matching the Puzzle path's Demazure-side accounting at $q = 0$.

(ii) $(1+q)^{\lfloor n/2 \rfloor}$ is the *coboundary / symmetriser* contribution—one factor of $(1+q)$ per odd-indexed generator $s_1, s_3, \ldots$ in the subgroup H. The exponent $\lfloor n/2 \rfloor$ is exactly the number of commuting $(1 + T_s)$ idempotent pairs in the Lattice path.

(iii) $p_\lambda(q)$ is the *Fock* contribution—palindromic, positive, and conjecturally a carrier of Kazhdan–Lusztig type positivity. We do not yet know what duality its palindromy reflects.

The factorisation is an *observation*, not a theorem: we have verified it by direct diagonalisation for all $\lambda \vdash n$ with $n \le 8$, but an algebraic proof is open. The natural route is to extract the $(1+q)^{\lfloor n/2 \rfloor}$ factor from the commuting structure of the odd generators, reducing the problem to understanding the residual p_λ. At the moment the factorisation is the shadow of a structure we have not yet named.

5.6 The Phase Diagram

We can now assemble the full picture. The categorical phase diagram is a two-dimensional array (see Figure 1) whose axes encode two independent structural variations:

Vertical axis (integrability level / categorical type): braided monoidal $\to$ coboundary monoidal $\to$ symmetric monoidal, governed by the Hecke chain $H_q \to H_0 \to \mathbb{C}[S_k]$. Each downward step forgets a layer of structure: first the spectral parameter, then the multiplicity bundle.

Horizontal axis (deformation type): classical (cohomological) $\to$ K-theoretic (trigonometric) $\to$ quantum K-theoretic (elliptic-like) $\to$ spin (higher-rank). Each rightward step deforms the R-matrix and the structure constants it computes, without changing the categorical type.

The filled and partially filled cells are:

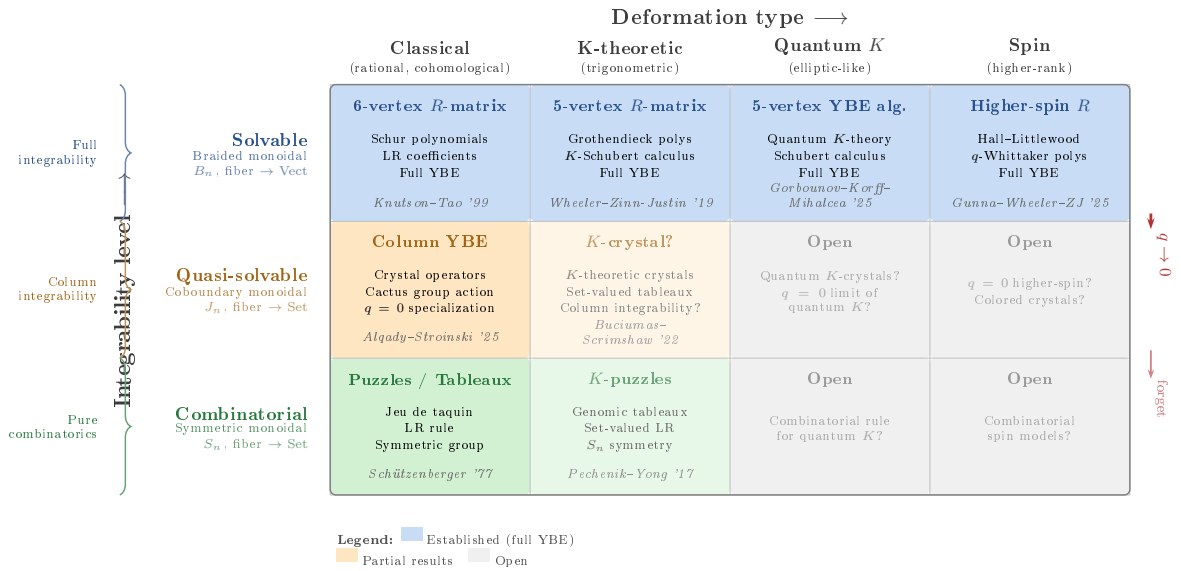

Figure 1: The categorical phase diagram. The vertical axis parameterises the integrability level (equivalently, the monoidal category type); the horizontal axis parameterises the deformation type. Filled cells have established lattice models; lightly shaded cells have partial results; empty cells are open. The downward arrows indicate the forgetful functors governed by the Hecke chain $H_q \to H_0 \to \mathbb{C}[S_k]$.

	Cohomological	**K-theoretic**	**Quantum K**	**Spin**
Braided	Schur / 6-vertex (rational)	Grothendieck / 5-vertex (trig.)	Quantum K-Schubert	Macdonald / spin HL
Coboundary	Key / colored 5-vertex [40]	*(partially explored)*		
Symmetric	monomials / trivial	*(partially explored)*		

The top row is densely populated: each entry has a solvable lattice model with a well-understood R-matrix and positive structure constants. The cohomological column is fully understood at all three levels. The remaining entries of the coboundary and symmetric rows at the K-theoretic, quantum K, and spin levels are largely open — filling them in is one of the central challenges for the field.

A third dimension?

The Tikhonov invariant $\kappa = (1 - qp)/(1 - p)$ suggests that the phase diagram may have more structure than a flat grid. The Möbius formula (24) foliates the (p, q) plane into curves of constant κ: each curve represents a one-parameter family of (p, q) values producing the same asymptotic behaviour. The cactus locus — the set of (p, q) values where the cactus generator emerges at $p = 1/2$ — traces the line $\kappa = 2 - q$, a constraint relating the interpolation and deformation parameters.

Question 5.22. Is the Möbius action on (p, q) related to the PGL(2) action on spectral parameters (Remark 5.12)? If so, κ would serve as a coordinate on a third axis — the *transition depth* between integrability levels — and the phase diagram would live on a surface rather than a plane, with vertical transitions mediated by a continuous invariant. One can test this by computing R-matrix eigenvalues and the Tikhonov invariant κ for small examples and checking

whether they transform under the same PGL(2).

What the diagram explains

We close by returning to the question that opened the paper: why do so many independent combinatorial models for LR coefficients exist?

The phase diagram provides an answer at two levels. *Horizontally*, models proliferate because the Yang-Baxter equation generates the Yang-Baxter groupoid (Section 4.1): every R-matrix move produces a new model computing the same coefficient. The groupoid is rich because the YBE has many solutions and many symmetries; the bijections between models are morphisms, not miracles. *Vertically*, models proliferate because the categorical hierarchy provides multiple projections of the same structure: the braided level sees the spectral parameter, the coboundary level sees the multiplicity bundle, and the symmetric level sees only the bare coefficient. Each level has its own characters, its own symmetry group, its own notion of integrability — and each level casts its own combinatorial shadow.

The coefficients themselves — $c^{\nu}_{\lambda,\mu}$, those nonnegative integers governing the multiplication of Schur functions — sit at the intersection of all projections. They are the invariant that every level computes, every model records, and every forgetful functor preserves. The multiplicity of descriptions is not a defect of the theory but a reflection of its depth: the same integers, seen from above through a spectral parameter, from the middle through a multiplicity bundle, and from below through a crystal basis, look different at each level but agree on the count. The phase diagram is the map that says which view you are taking and what you have chosen not to see.

6 The Shuffle Algebra Substrate

The three-level hierarchy of Section 5 organises the space of LR models vertically (by integrability level) and horizontally (by deformation type). But the hierarchy itself sits inside a larger algebraic universe. In this section we sketch the contours of that universe: the *shuffle algebra* framework, which provides a single algebraic substrate from which all the integrable vertex models of this paper can be derived.

6.1 Feigin-Odesskii shuffle algebras

Let V be a finite-dimensional vector space and let $\zeta(z) \in \mathbb{C}(z)$ be a rational function, called the *shuffle kernel*. The *Feigin-Odesskii shuffle algebra* [**?**] is defined on the direct sum

$$\mathcal{S}_\zeta \;=\; \bigoplus_{n\geq 0} \mathbb{C}(z_1,\ldots,z_n),$$

with multiplication given by the *shuffle product*: for $f \in \mathbb{C}(z_1,\ldots,z_m)$ and $g \in \mathbb{C}(z_1,\ldots,z_n)$,

$$(f\star g)(z_1,\ldots,z_{m+n}) \;=\; \mathrm{Sym}\left[f(z_1,\ldots,z_m)\cdot g(z_{m+1},\ldots,z_{m+n})\cdot \prod_{i=1}^{m}\prod_{j=m+1}^{m+n}\zeta\left(\frac{z_i}{z_j}\right)\right], \tag{25}$$

where Sym denotes symmetrisation over all (m,n)-shuffles of the variables $z_1,\ldots,z_{m+n}$. The shuffle kernel ζ controls the algebra: different choices of ζ produce genuinely different algebraic structures. At the extreme $\zeta \equiv 1$, the shuffle product reduces to ordinary symmetrisation and $\mathcal{S}_\zeta$ is the algebra of symmetric rational functions. Nontrivial kernels introduce interaction between the two factors, and it is precisely this interaction that encodes the Boltzmann weights of integrable lattice models.

6.2 From shuffle algebras to vertex models

The connection to our story was established by Garbali and Gunna [19], who proved that the vertex models computing symmetric function structure constants—including the Schur, Grothendieck, and Hall-Littlewood models of Section 3—can be realised as *representations* of appropriate shuffle algebras. More precisely, the row operators of the lattice model (the transfer matrices $T(u)$ of Section **??**) intertwine the shuffle product with the composition of vertex weights: the algebraic identity $T(u) \star T(v) = T(u) \circ T(v)$ (suitably interpreted) encodes both the Yang-Baxter equation and the commutativity of transfer matrices in a single stroke.

Garbali and Negut [20] extended this picture by proving that the shuffle algebra $\mathcal{S}_\zeta$, for the kernel $\zeta(z) = (z-1)/(z-q)$ relevant to the Schur/Hall-Littlewood family, is isomorphic to the positive part $U_q^+(\widehat{\mathfrak{gl}}_1)$ of the quantum toroidal algebra (equivalently, the positive half of the Yangian $Y(\widehat{\mathfrak{gl}}_1)$ at the rational level). This isomorphism is not merely formal: it identifies the shuffle generators with specific combinations of Drinfeld currents, and under this identification, the vertex model partition functions become matrix elements of intertwining operators between Fock space modules. The consequence is stark: *all* the integrable lattice models from Section 3—Schur, Grothendieck, Hall-Littlewood, q-Whittaker—are manifestations of a single algebraic structure, viewed through different representations.

6.3 Connection to the Hecke algebra

The shuffle algebra also makes contact with the Hecke transition algebra of Section 5. Kalmykov [25] established explicit isomorphisms connecting three objects at the rational level:

1. the Feigin-Odesskii shuffle algebra $\mathcal{S}_\zeta$;
2. the *degenerate affine Hecke algebra* (dAHA) H_n^{aff}, which extends the finite Hecke algebra $H_q(S_n)$ by polynomial generators $x_1, \ldots, x_n$ satisfying the Bernstein-Lusztig relations; and
3. the Yangian $Y(\mathfrak{gl}_n)$.

Since the finite Hecke algebra $H_q(S_n)$ embeds in the dAHA as the subalgebra generated by the simple transpositions (without the polynomial generators), the Hecke transition algebra from Section 5 sits inside the shuffle substrate as a natural subalgebra. The three-level hierarchy $H_q \to H_0 \to \mathbb{C}[S_n]$ is a shadow of a deeper hierarchy within the Yangian, where the specialisation $q \to 0$ corresponds to a contraction of the quantum group to its classical limit.

6.4 The emerging picture

The shuffle algebra framework suggests a four-level depth hierarchy for understanding why the LR coefficients admit so many combinatorial models:

1. **Combinatorial bijections.** At the surface, one has explicit maps between tableaux, puzzles, pipe dreams, and other models. These bijections are typically proved by induction or sign-reversing involutions, and their existence appears miraculous.
2. **Yang-Baxter groupoid.** One level down, the bijections are morphisms in the Yang-Baxter groupoid of Bump and Naprienko [11] (Section 4.1). Their existence follows from the Yang-Baxter equation: each R-matrix move preserves the partition function, so composing moves gives a canonical bijection. The miracle is demystified but not fully explained—one still needs to know *why* the R-matrix satisfies the YBE.

3. **Shuffle algebra.** At the next level, the R-matrices are intertwining operators for representations of the shuffle algebra $\mathcal{S}_\zeta$. The YBE is a consequence of the associativity of the shuffle product, and the multiplicity of lattice models reflects the multiplicity of representations. Different representations give different vertex weights, but the partition functions agree because the underlying algebra is the same.

4. **Geometric Satake.** At the deepest level (which we do not develop here), the shuffle algebra is the cohomology of a moduli space of instantons, and the LR coefficients are intersection numbers on the affine Grassmannian. The geometric Satake correspondence [**?**] provides the ultimate explanation: the structure constants of the representation ring are geometric invariants, and their positivity is a consequence of the decomposition theorem in perverse sheaves.

Each level subsumes the ones above it, offering a deeper explanation for the same numerical coincidences. The categorical phase diagram of Section 5 lives at levels 2 and 3: the vertical axis (integrability level) is governed by the Hecke quotients of the shuffle algebra, while the horizontal axis (deformation type) is governed by the choice of shuffle kernel ζ. The full picture—a phase diagram sitting inside the representation theory of a shuffle algebra, itself a shadow of geometric Satake—remains to be completed. We turn to the open questions in the next section.

7 Open Questions and Future Directions

We organise the questions suggested by this paper into three tiers, roughly ordered by accessibility: problems that could be attacked with existing tools, problems that require new ideas, and problems that call for significant new theory.

7.1 Immediately tractable

Q1. Complete the phase diagram horizontally. The categorical phase diagram of Section 5 has populated cells in every column of the solvable row (Schur, Grothendieck, quantum K, Hall-Littlewood) and in the classical column of the coboundary row (Demazure crystals). The remaining coboundary cells—K-theoretic, quantum K-theoretic, and spin models at the quasi-solvable level—are open. Yang's Demazure crystal model [40] covers the classical coboundary column; the K-theoretic and spin columns at this level remain to be constructed. Concretely: which lattice models compute coboundary-level structure constants for Grothendieck polynomials and Hall-Littlewood polynomials? Do they exhibit the factored (column-level) YBE characteristic of quasi-solvability?

Q2. Type C/D phase diagram. Everything in this paper concerns type A—the general linear group and its symmetric functions. Azenhas, Feller, and Torres [5] construct the symplectic cactus group action on type C crystals, discovering additional relations beyond those present in type A. What does the integrability hierarchy look like for Sp_{2n}? The quasi-solvable lattice models of Buciumas and Scrimshaw [10] for types B, C, and D provide the integrable side of the story; the categorical side—what replaces the cactus group, whether the Hecke transition algebra admits a type C analogue—is largely unexplored.

Q3. J_n non-splitting for all n. The cactus group J_n fits into a short exact sequence $1 \to PJ_n \to J_n \to S_n \to 1$, where PJ_n is the *pure cactus group*. For $n = 3$, we proved by four independent methods (including the computation $H^2(S_3, \mathbb{Z}^{\mathrm{sgn}}) \cong \mathbb{Z}/3\mathbb{Z}$) that this sequence does not split. Non-splitting is the algebraic counterpart of the geometric fact that the real moduli space $\overline{M}_{0,n+1}(\mathbb{R})$ is not a trivial S_n-bundle.

For $n \geq 4$, the pure cactus group is non-abelian, so the classical group cohomology approach (which requires abelian coefficients) does not directly apply. The most promising route is topological: Devyatov's presentation [12] of the fundamental group of $\overline{M}_{0,n+1}(\mathbb{R})$ expresses the pure cactus group in terms of explicit generators and relations. A careful analysis of these relations—perhaps combined with the non-abelian cohomology $H^1(S_n, \mathrm{Out}(PJ_n))$—should settle the non-splitting question for all n.

Q4. The functor puzzle behind the H-invariant theorem. Theorem 5.14—the identity $r_\lambda = \dim V_\lambda^H$ for $H = \langle s_1, s_3, s_5, \ldots \rangle \cong S_2^{\lfloor n/2 \rfloor}$—is now established [2]. The proof unifies three specializations of the same staircase operator: at $q = 0$ the product Π_0 is Fomin–Stanley's Schubert-polynomial generator on the 0-Hecke algebra; at $q = 1$ it becomes the classical symmetric-group staircase, whose rank is $\dim V_\lambda^H$ by a direct Frobenius argument; for generic $q > 0$, eigenvalue positivity—a consequence of Kazhdan–Lusztig W-graph non-negativity—locks the rank to the $q = 1$ value. Three windows on one operator, a single rank throughout.

What remains open is *categorical*. The theorem equates two dimensions, but the underlying subspaces are not the same: $\mathrm{im}(\Pi|_{V_\lambda}) \neq V_\lambda^H$ in general (Remark 5.16). The functors $V_\lambda \mapsto \mathrm{im}(\Pi|_{V_\lambda})$ and $V_\lambda \mapsto V_\lambda^H$ agree on dimensions in every irrep yet differ as subfunctors of the identity. Is there a natural isomorphism between them? Such an isomorphism would be a genuine representation-theoretic bridge between the projection picture (image of a Hecke idempotent) and the invariant-theory picture (fixed points of a Young subgroup), and would likely connect to a known categorical construction in the Schur–Weyl or descent-algebra world.

Q5. Key positivity from coboundary structure. Alqady and Stroinski [4] proved that TL_0—the Temperley-Lieb quotient of the 0-Hecke algebra—carries a coboundary monoidal structure. Independently, Paten and Woodruff [33] showed that key polynomials (Demazure characters) are the H_0-characters, and that certain TL immanants are key-positive. Do these two facts compose? If the coboundary commutor on TL_0 preserves positivity in the key basis, this would give a categorical explanation for key positivity of Schur expansions—the most sought-after structural result in the Demazure character theory.

7.2 Medium-term

Q6. κ as spectral parameter. Tikhonov [38] showed that the limiting permuton of a random sorting network is governed by the invariant $\kappa = (1 - qp)/(1 - p)$, and that the Möbius transformation $p' = p(1-q)/(1-qp)$ relates the braided ($q > 0$) and coboundary ($q = 0$) regimes. This transformation is an element of $\mathrm{PGL}(2)$, acting on the probability parameter p with the Hecke parameter q as coefficient.

The same group $\mathrm{PGL}(2)$ acts on the spectral parameter u in the Yang-Baxter equation: the R-matrix $R(u)$ of a rational model transforms under $u \mapsto (au + b)/(cu + d)$ by conjugation. Is Tikhonov's Möbius transformation the *same* $\mathrm{PGL}(2)$ element that relates the spectral parameters of the braided and coboundary R-matrices? An affirmative answer would give a quantitative, parameter-level avatar of the forgetful functor Braided $\to$ Coboundary from Section 5.

Q7. Cactus action on Yang's boundary conditions. Yang's Demazure crystal model [40] is parameterised by a permutation $w \in S_r$, which enters as the boundary condition on the right edge of the lattice. The cactus group J_r acts on Demazure crystals via compositions of Schützenberger involutions [23]. Does the cactus action on the crystal correspond to a permutation of Yang's boundary conditions—that is, does the cactus generator $s_{[i,j]}$

send the boundary condition w to $w \cdot w_0^{[i,j]}$, where $w_0^{[i,j]}$ is the longest element of $S_{\{i,\dots,j\}}$? If so, the coboundary commutor would have an explicit lattice-model manifestation: it permutes the boundary data, and the resulting bijection on configurations is the lattice-model avatar of the Schützenberger involution.

Q8. Specialization moduli. Fan, Guo, and Xiong [13] proved that KTW puzzles and bumpless pipe dreams are not independent models but rather *specialisations* of a single lattice model: different choices of boundary conditions and spectral parameters in one unified vertex model recover both combinatorial objects. What is the *moduli space* of all such specialisations? Is it related to the space of factored integrability conditions (the locus where R_{row} and R_{col} independently satisfy the YBE)? A complete description of this moduli space would answer, in a precise sense, the question "how many essentially different combinatorial models for the LR coefficients exist?"

7.3 Long-term

Q9. Shuffle algebra for the full phase diagram. The Garbali-Negut isomorphism [20] connects shuffle algebras to Yangians at the rational level, which governs the classical (cohomological) column of the phase diagram. Does this isomorphism extend to the trigonometric level (K-theoretic column, where the relevant quantum group is the quantum affine algebra $U_q(\widehat{\mathfrak{sl}}_n)$) and the elliptic level (where the quantum group is Felder's elliptic quantum group [14])? The Shibukawa-Ueno hierarchy [36]—elliptic, trigonometric, rational—classifies R-matrices by the function field in which they live, and is *orthogonal* to the braided/coboundary/symmetric hierarchy of Section 5. The intersection of these two hierarchies would give a three-dimensional phase diagram: one axis for deformation type, one for integrability level, and one for the analytic class of the R-matrix.

Q10. TQFT for puzzle bundles. The failure of the Zinn-Justin tetrahedron equation at the quasi-solvable level (Section 4.3) defines a curvature 2-form on the "puzzle bundle" over the higher Bruhat order $B(n, 2)$. This curvature measures the obstruction to promoting column-level integrability to full vertex-level integrability, and its holonomy around closed loops encodes the non-abelian structure of the pure cactus group. Is there a topological quantum field theory whose partition function computes LR coefficients, with the curvature encoding the quasi-solvable level? Such a TQFT would provide a geometric home for the categorical phase diagram: the braided level would correspond to a flat connection (zero curvature, trivial holonomy), the coboundary level to a curved connection with discrete holonomy, and the symmetric level to the trivial bundle.

Q11. KPZ universality for Schubert calculus. Morales, Panova, Petrov, and Yeliussizov [31] connect pipe dreams to the totally asymmetric simple exclusion process (TASEP) and Tracy-Widom distributions, placing classical Schubert calculus in the KPZ universality class. Does this connection extend to K-theoretic or quantum K-theoretic Schubert calculus? The K-theoretic structure constants are *signed*, which complicates the probabilistic interpretation, but Tikhonov's permuton results [38] suggest that the probabilistic side is tractable whenever the Hecke parameter q is real. A KPZ universality theorem for Grothendieck polynomials would link the algebraic geometry of flag varieties to the statistical mechanics of interface growth—two subjects that have developed largely in parallel.

Coda. The central message of this paper is that the multiplicity of combinatorial models for LR coefficients is not a historical accident but a structural inevitability. The models proliferate because they are representations of a rich algebraic object—the shuffle algebra and its

quotients—organised by the integrability hierarchy into levels of increasing symmetry. Each level forgets something precise (a spectral parameter, a multiplicity bundle), and each forgetting produces a simpler but still faithful encoding of the same nonnegative integers. Understanding this structure does more than explain existing combinatorics: it predicts where new models should live (the open cells of the phase diagram), identifies the invariants that distinguish them (the Hecke parameter q, the shuffle kernel ζ), and points toward deeper explanations (geometric Satake, TQFT, KPZ universality) that may one day subsume the entire picture.

The LR coefficients, those modest nonnegative integers governing the multiplication of Schur functions, have turned out to be a window onto a remarkably deep landscape. We have tried to draw a map of what we can see. The territory beyond the map—the open cells, the conjectural connections, the questions we do not yet know how to ask—is, we suspect, richer still.

Acknowledgements

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